



Microsoft Sentinel Training Lab

The official open-source **Microsoft Sentinel Training Lab** is the best starting point for hands-on TI learning.

- Repo: <https://github.com/Azure/Azure-Sentinel/blob/master/Tools/Microsoft-Sentinel-Training-Lab/README.md>
- Blog: <https://techcommunity.microsoft.com/blog/microsoftsentinelblog/introducing-the-microsoft-sentinel-training-lab-hands-on/4513274>

Onboarding — Setting up the environment

Create a User-Assigned Managed Identity (UAMI)

Automated setup (Azure Cloud Shell Bash)

The setup script creates or reuses `SentinelDetectionRulesIdentity`, waits for the managed identity service principal to become available in Microsoft Graph, grants and verifies `CustomDetection.ReadWrite.All`, and prints the resource ID required by the deployment template.

Open Azure Cloud Shell, select **Bash**, and run:

```
curl -fsSL sentinel-training-lab-onboarding.sh \
  https://raw.githubusercontent.com/Azure/Azure-Sentinel/master/Tools/Microsoft-Sentinel-Training-Lab/Tools/sentinel-training-lab-onboarding.sh

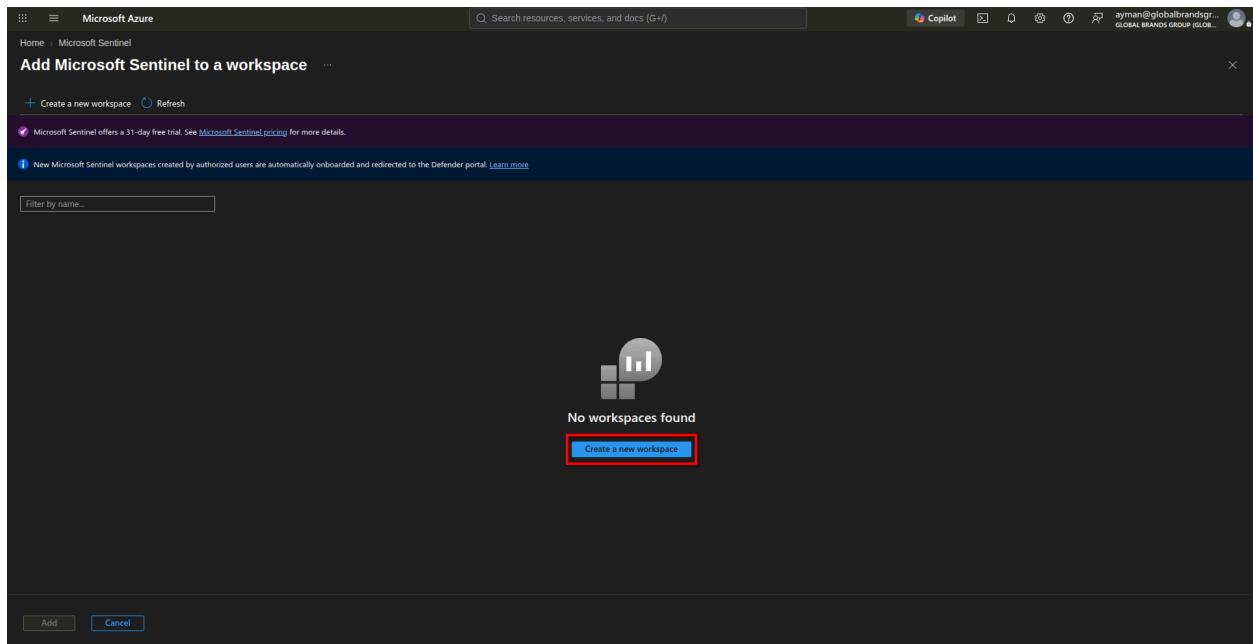
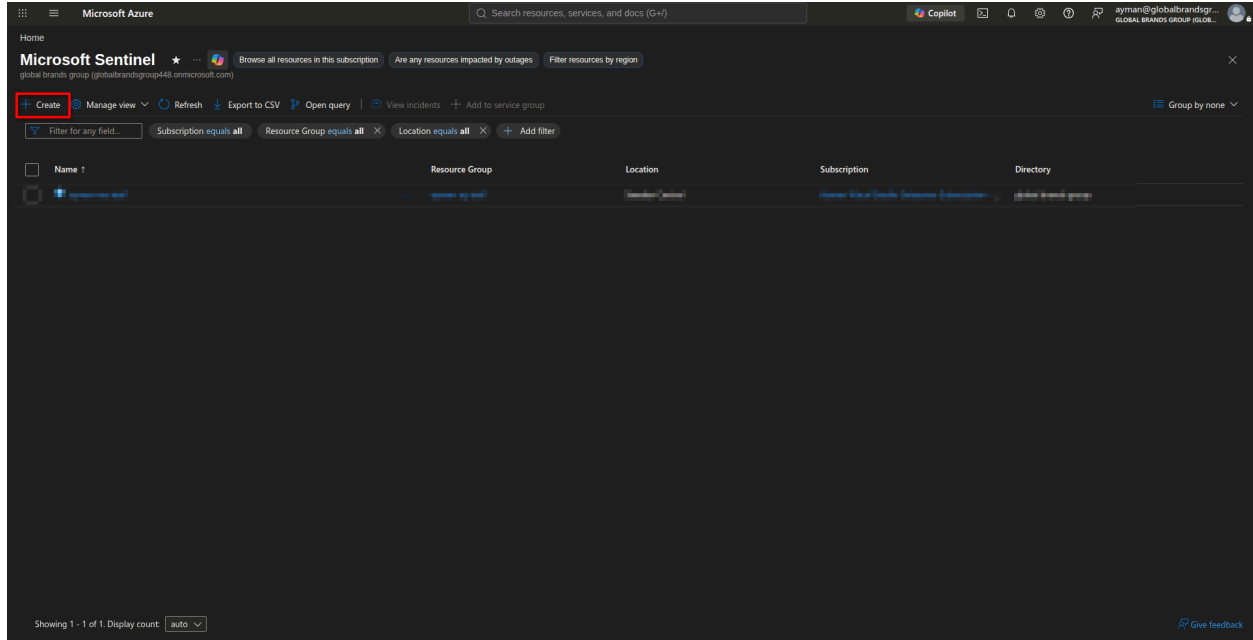
bash sentinel-training-lab-onboarding.sh <your-resource-group>
```

Name	Subscription	Location
ayman-Microsoft-Foundry	Ayman-Visual Studio Enterprise Subscription - MPN	Sweden Central
ayman-rg-test1	Ayman-Visual Studio Enterprise Subscription - MPN	Sweden Central
MicrosoftSentinelTrainingLab	Ayman-Visual Studio Enterprise Subscription - MPN	Sweden Central
NetworkWatcherRG	Ayman-Visual Studio Enterprise Subscription - MPN	UAE North

```
PS /home/ayman> curl -fsSL sentinel-training-lab-onboarding.sh https://raw.githubusercontent.com/Azure/Azure-Sentinel/master/Tools/Microsoft-Sentinel-Training-Lab/Tools/sentinel-training-lab-onboarding.sh
PS /home/ayman> bash sentinel-training-lab-onboarding.sh MicrosoftSentinelTrainingLab
[*] Azure context
Subscription: ayman-visual-studio-enterprise-subscription-mpn
Tenant: ayman-visual-studio-enterprise-subscription-mpn
[*] Reusing UAMI 'SentinelDetectionRulesIdentity'.
[!] Tenant mismatch.
Active tenant: ayman-visual-studio-enterprise-subscription-mpn
Identity tenant: ayman-visual-studio-enterprise-subscription-mpn
PS /home/ayman>
```

Create a Log Analytics workspace

1. In the [Azure portal](#), search for **Sentinel** and select **Microsoft Sentinel**.
2. Select **Create** → **Create a new workspace**.
3. Choose your **Subscription**, **Resource Group**, **Workspace Name**, and **Region**.
4. Select **Review + Create**, then **Create**.



Microsoft Azure

Home > Microsoft Sentinel > Add Microsoft Sentinel to a workspace

Create Log Analytics workspace

Basics Tags Review & Create

Basics

A Log Analytics workspace is the basic management unit of Azure Monitor Logs. There are specific considerations you should take when creating a new Log Analytics workspace. [Learn more](#)

With Azure Monitor Logs you can easily store, retain, and query data collected from your monitored resources in Azure and other environments for valuable insights. A Log Analytics workspace is the logical storage unit where your log data is collected and stored.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription

Resource group

Instance details

Name

Region

Review & Create Previous Next > Tags >

Add Microsoft Sentinel to your workspace

1. Select **Create**.

Microsoft Azure

Home

Microsoft Sentinel

global brands group (globalbrandsgr448.onmicrosoft.com)

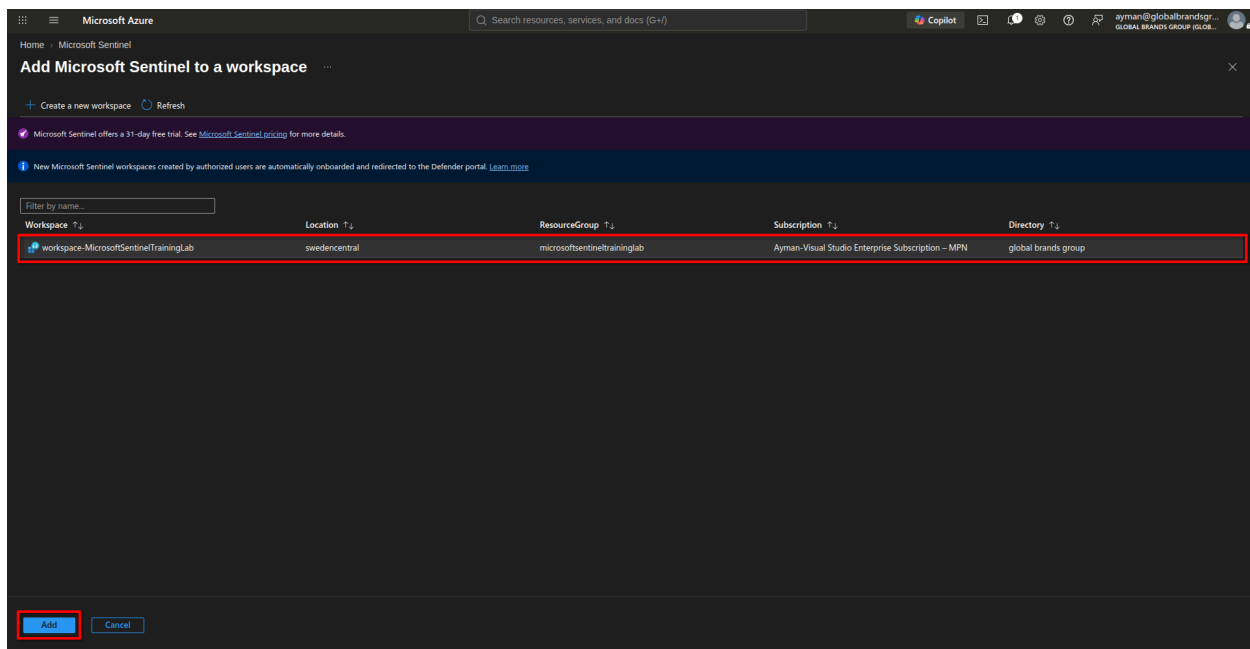
Create Manage view Refresh Export to CSV Open query View incidents Add to service group

Filter for any field Subscription equals all Resource Group equals all Location equals all Add filter

Name	Resource Group	Location	Subscription	Directory
workspace-MicrosoftSentinelTrainingLab	MicrosoftSentinelTrainingLab	Sweden Central	Ayman-Visual Studio Enterprise Subscription - MPN	global brands group

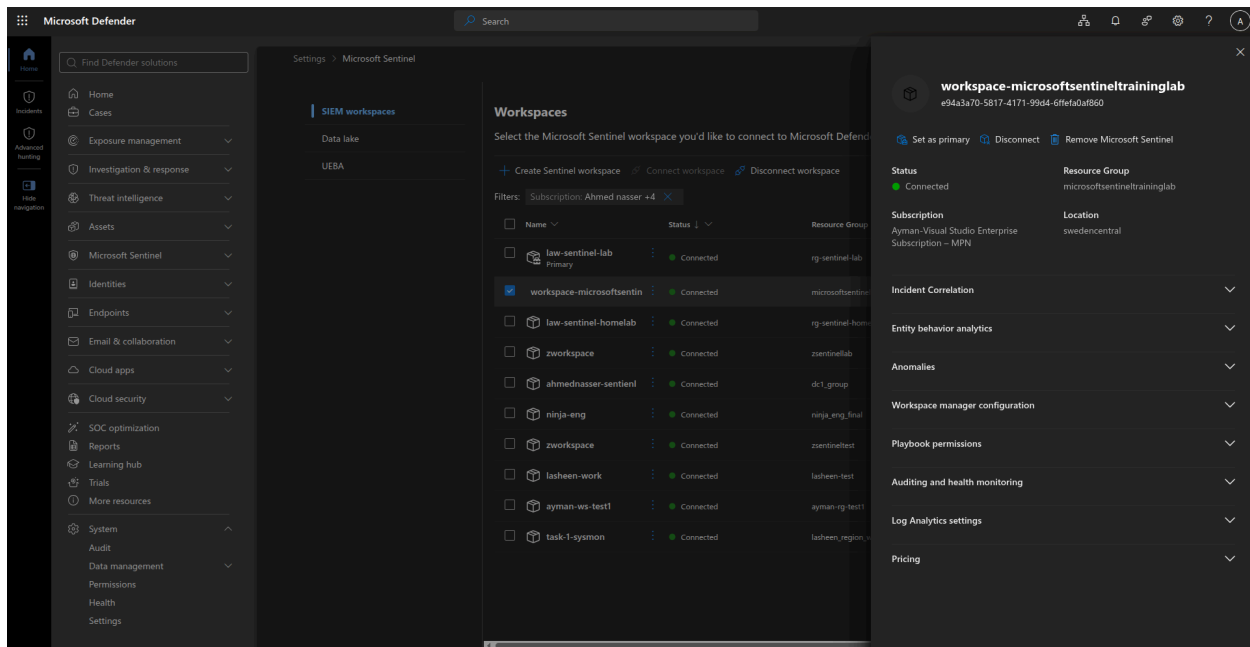
Showing 1 - 1 of 1. Display count: auto

2. Select the workspace you created in Step 1 and select **Add**.



Access Microsoft Sentinel in the Defender portal

1. Go to **Settings > Microsoft Sentinel > Workspaces**, verify your workspace is listed, and click **Connect**.



Deploy the Microsoft Sentinel Training Lab Solution

This deploys pre-recorded telemetry (~20 MB) and creates analytics rules, workbooks, watchlists, and playbooks used in the subsequent exercises.

Make sure you have completed the **Custom Detection Rules Setup** from the [README](#) (creating the User-Assigned Managed Identity). You will need the UAMI resource ID during deployment.

<https://portal.azure.com/#create/Microsoft.Template/uri/https%3A%2F%2Fraw.githubusercontent.com%2FAzure%2FAzure-Sentinel%2Fmaster%2FTools%2FMicrosoft-Sentinel-Training-Lab%2FPackage%2FmainTemplate.json>

1. Select the **Subscription**, **Resource Group**, and **Workspace** from the previous steps. Make sure the **Region** matches the region of your existing workspace.
2. Under **Detection Rules Identity Resource Id**, paste the full resource ID of your UAMI (or leave empty to skip detection rules deployment).
3. Select **Review + create**, then **Create**.

Note: The official ARM template above is outdated — it cannot onboard Sentinel to the workspace and needs a delay (Sentinel onboarding time) for the deployment to succeed. The corrected template below was used for this lab; it enables Sentinel onboarding on the workspace and waits for it to propagate before deploying the remaining resources.

Microsoft Sentinel-enabled ARM template

```
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.js
on#",
  "contentVersion": "1.0.6",
  "parameters": {
    "location": {
      "type": "string",
      "minLength": 1,
      "defaultValue": "[resourceGroup().location]",
      "metadata": {
        "description": "Not used, but neede
d to pass arm-ttk test `Location-Should-Not-Be-Hardcoded`. We instead use the `workspace-lo
cation` which is derived from the LA workspace"
      }
    },
    "workspace-location": {
      "type": "string",
      "minLength": 1,
      "defaultValue": "[parameters('locatio
n')]",
      "metadata": {
        "description": "Region to
deploy solution resources"
      }
    },
    "workspaceName": {
      "type": "string",
      "metadata": {
        "description": "The Log Analyt
ics workspace name (4-63 letters, digits or '-'). The '-' shouldn't be the first or the last
symbol"
      }
    },
    "formattedTimeNow": {
      "type": "string",
      "defaultValue": "[utcNow('g')]",
      "metadata": {
        "description": "Appended to
workbook displayName to make them unique"
      }
    }
  }
}
```

```

    },
    "artifactsVersion": {
      "type": "string",
      "defaultValue": "[utcNow('yyyyMMddHHmmss')]",
      "metadata": {
        "description": "Cache-busting version for linked templates"
      }
    },
    "scheduleStartTime": {
      "type": "string",
      "defaultValue": "[dateTimeAdd(utcNow('yyyy-MM-ddTHH:mm:ssZ'), 'PT15M')]",
      "metadata": {
        "description": "UTC start time for the one-time ingestion schedule"
      }
    },
    "detectionRulesScheduleStartTime": {
      "type": "string",
      "defaultValue": "[dateTimeAdd(utcNow('yyyy-MM-ddTHH:mm:ssZ'), 'PT30M')]",
      "metadata": {
        "description": "UTC start time for the detection-rules runbook (runs after ingestion)"
      }
    },
    "detectionRulesIdentityResourceId": {
      "type": "string",
      "defaultValue": "",
      "metadata": {
        "description": "Full resource ID of a pre-created User-Assigned Managed Identity with Microsoft Graph CustomDetection.ReadWrite.All permission. Leave empty to skip detection rules deployment."
      }
    },
    "enableIngestion": {
      "type": "bool",
      "defaultValue": true,
      "metadata": {
        "description": "Whether to run the ingestEvents deployment script. Disabled for tenants that block storage account key authentication."
      }
    },
    "sentinelOnboardingWaitSeconds": {
      "type": "int",
      "defaultValue": 300,
      "minValue": 60,
      "maxValue": 600,
      "metadata": {
        "description": "Sentinel onboarding wait time in seconds"
      }
    }
  }
}

```

```

n": "Seconds to wait for Sentinel onboarding to propagate before deploying Sentinel resource
s."

        },
        "workbook1-name": {
            "type": "string",
            "defaultValue": "LabWorkbook",
            "metadata": {
                "description": "Name for the
workbook"
            }
        },
        "playbook1-PlaybookName": {
            "defaultValue": "Get-GeoFromIpAndTagIn
cident",
            "type": "string",
            "minLength": 1,
            "metadata": {
                "description": "Resou
rce name for the logic app playbook. No spaces are allowed"
            }
        }
    },
    "variables": {
        "_artifactsLocation": "https://raw.githubusercontent.com/Azure/Azure-S
entinel/master/Tools/Microsoft-Sentinel-Training-Lab/",
        "automationAccountName": "[concat('sentineltraining', uniqueString(res
ourceGroup().id))]",
        "scheduleName": "[concat('IngestTrainingTelemetryOnce-', uniqueString
(resourceGroup().id, deployment().name))]"
    },
    "resources": [
        {
            "name": "workspaceCreation",
            "type": "Microsoft.Resources/deployments",
            "apiVersion": "2024-07-01",
            "properties": {
                "mode": "Incremental",
                "templateLink": {
                    "uri": "[uri(variables('_a
rtifactsLocation'), concat('Artifacts/LinkedTemplates/workspace.json?v=', parameters('artifac
tsVersion')))]",
                    "contentVersion": "1.0.0.
0"
                }
            },
            "parameters": {
                "workspaceName": {
                    "valu
e": "[parameters('workspaceName')]"
                },
                "pricingTier": {
                    "value":
"PerGB2018"
                }
            },

```

```

        "dataRetention": {
            "value": 90
        },
        "location": {
            "value": "[parameters('workspace-location')]"
        }
    },
    {
        "name": "default",
        "type": "Microsoft.SecurityInsights/onboardingStates",
        "apiVersion": "2024-03-01",
        "scope": "[resourceId('Microsoft.OperationalInsights/workspaces',
parameters('workspaceName'))]",
        "dependsOn": [
            "[resourceId('Microsoft.Resources/deployments',
'workspaceCreation')]"
        ],
        "properties": {
            "customerManagedKey": false
        }
    },
    {
        "name": "waitForSentinelOnboarding",
        "type": "Microsoft.Resources/deploymentScripts",
        "apiVersion": "2023-08-01",
        "location": "[parameters('workspace-location')]",
        "kind": "AzureCLI",
        "dependsOn": [
            "[extensionResourceId(resourceId('Microsoft.OperationalInsights/workspaces', parameters('workspaceName')), 'Microsoft.SecurityInsights/onboardingStates', 'default')]"
        ],
        "properties": {
            "forceUpdateTag": "[parameters('artifactsVersion')]",
            "azCliVersion": "2.57.0",
            "arguments": "[string(parameters('sentinelOnboardingWaitSeconds'))]",
            "scriptContent": "set -eu\nsleep \"$1\"\nprintf '{\"waited\":true}' > \"\$AZ_SCRIPTS_OUTPUT_PATH\"",
            "timeout": "PT15M",
            "cleanupPreference": "Always",
            "retentionInterval": "P1D"
        }
    },
    {
        "name": "deployDetectionRules",
        "type": "Microsoft.Resources/deployments",
        "apiVersion": "2024-07-01",
        "condition": "[not(empty(parameters('detectionRulesIdentityResource

```

```

eId')))]]",
    "dependsOn": [
      "[resourceId('Microsoft.Resources/deployments',
'workspaceCreation')]"
    ],
    "properties": {
      "mode": "Incremental",
      "templateLink": {
        "uri": "[uri(variables('_a
rtifactsLocation'), concat('Artifacts/LinkedTemplates/deployDetectionRules.json?v=', paramete
rs('artifactsVersion')))]",
        "contentVersion": "1.0.0.
0"
      },
      "parameters": {
        "scheduleStartTime": {
          "va
lue": "[parameters('detectionRulesScheduleStartTime')]"
        },
        "userAssignedIdentityResource
Id": {
          "value": "[parameters('detectionRulesIdentityResourceId')]"
        }
      }
    }
  },
  {
    "name": "ingestEvents",
    "type": "Microsoft.Resources/deployments",
    "apiVersion": "2024-07-01",
    "condition": "[parameters('enableIngestion')]",
    "dependsOn": [
      "[resourceId('Microsoft.Resources/deployments',
'workspaceCreation')]"
    ],
    "properties": {
      "mode": "Incremental",
      "expressionEvaluationOptions": {
        "scope": "I
nner"
      },
      "template": {
        "$schema": "https://schema.man
agement.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
        "contentVersion": "1.0.0.0",
        "parameters": {
          "WorkspaceNa
me": {
            "type": "string",
            "metadata": {

```

```

    "description": "Log Analytics workspace name"
  },
  {
    "type": "string",
    "metadata": {
      "description": "UTC start time for the one-time schedule"
    },
    "scheduleStart": "2020-01-01T00:00:00Z",
    "scheduleName": "Log Analytics workspace name",
    "location": "Log Analytics workspace name",
    "functions": [
      {
        "name": "Log Analytics workspace name",
        "variables": {
          "automationAccountName": "[concat('sentineltraining', uniqueString(resourceGroup().id))]"
        },
        "resources": [
          {
            "type": "Microsoft.Automation/automationAccounts",

```

```

on": "2024-10-23",
"[variables('automationAccountName')]",
n": "[parameters('location')]",
y": {
  "type": "SystemAssigned"
},
es": {
  "sku": {
    "name": "Basic"
  }
}
},
{
  "type":
    "Microsoft.Automation/automationAccounts/runbooks",
    "apiVersi
    "name":
    "locatio
    "depends0
on": "2024-10-23",
"[concat(variables('automationAccountName'), '/IngestTrainingTelemetry')]",
n": "[parameters('location')]",
n": [
  "[resourceId('Microsoft.Automation/automationAccounts', variables('automationAccountName'))]"
],
es": {
  "runbookType": "PowerShell",
  "logProgress": true,
  "logVerbose": true,
  "publishContentLink": {
    "uri": "[concat('https://raw.githubusercontent.com/Azure/Azure-Sentinel/master/Tools/Microso
ft-Sentinel-Training-Lab/Artifacts/Scripts/RunIngest.ps1', '?v=', parameters('scheduleNam
e'))]"
  }
}

```

```

    },
    {
      "type":
      "Microsoft.Automation/automationAccounts/schedules",
      "apiVersi
      "name":
      "depends0
      n": [
        "[resourceId('Microsoft.Automation/automationAccounts', variables('automationAccountName'))]"
      ],
      "properti
      es": {
        "frequency": "OneTime",
        "startTime": "[parameters('scheduleStartTime')]",
        "timeZone": "UTC"
      }
    },
    {
      "type":
      "Microsoft.Automation/automationAccounts/variables",
      "apiVersi
      "name":
      "depends0
      n": [
        "[resourceId('Microsoft.Automation/automationAccounts', variables('automationAccountName'))]"
      ],
      "properti
      es": {
        "isEncrypted": false,
        "value": "[concat('\'", subscription().subscriptionId, '\"])"
      }
    },
    {
      "type":
      "Microsoft.Automation/automationAccounts/variables",
      "apiVersi
      "name":
      "depends0
      n": [
        "[resourceId('Microsoft.Automation/automationAccounts', variables('automationAccountName'))]"
      ],
      "properti
      es": {
        "frequency": "OneTime",
        "startTime": "[parameters('scheduleStartTime')]",
        "timeZone": "UTC"
      }
    }
  ],
  "properties": {
    "frequency": "OneTime",
    "startTime": "[parameters('scheduleStartTime')]",
    "timeZone": "UTC"
  }
}

```



```

    "name":
    "[concat(variables('automationAccountName'), '/SentinelTrainingResourceGroupName')]",
    "dependsOn": [
    "[resourceId('Microsoft.Automation/automationAccounts', variables('automationAccountName'))]"
    ],
    "properties": {
    "isEncrypted": false,
    "value": "[concat('\'", resourceGroup().name, '\']]"
    }
  },
  {
    "type":
    "Microsoft.Automation/automationAccounts/variables",
    "apiVersion":
    "2024-10-23",
    "name":
    "[concat(variables('automationAccountName'), '/SentinelTrainingLocation')]",
    "dependsOn": [
    "[resourceId('Microsoft.Automation/automationAccounts', variables('automationAccountName'))]"
    ],
    "properties": {
    "isEncrypted": false,
    "value": "[concat('\'", parameters('location'), '\']]"
    }
  },
  {
    "type":
    "Microsoft.Automation/automationAccounts/variables",
    "apiVersion":
    "2024-10-23",
    "name":
    "[concat(variables('automationAccountName'), '/SentinelTrainingWorkspaceName')]",
    "dependsOn": [
    "[resourceId('Microsoft.Automation/automationAccounts', variables('automationAccountName'))]"
    ],
    "properties": {

```

```

    "isEncrypted": false,

    "value": "[concat('\'', parameters('WorkspaceName'), '\']')]"
  }

  },
  {
    "type":
    "Microsoft.Automation/automationAccounts/jobSchedules",
    "apiVersion":
    "2024-10-23",
    "name":
    "[concat(variables('automationAccountName'), '/', guid(variables('automationAccountName'), 'IngestTrainingTelemetry', parameters('scheduleName')))]",
    "dependsOn": [
      "[resourceId('Microsoft.Automation/automationAccounts/runbooks', variables('automationAccountName'), 'IngestTrainingTelemetry')]",
      "[resourceId('Microsoft.Automation/automationAccounts/schedules', variables('automationAccountName'), parameters('scheduleName'))]",
      "[resourceId('Microsoft.Automation/automationAccounts/variables', variables('automationAccountName'), 'SentinelTrainingSubscriptionId')]",
      "[resourceId('Microsoft.Automation/automationAccounts/variables', variables('automationAccountName'), 'SentinelTrainingResourceGroupName')]",
      "[resourceId('Microsoft.Automation/automationAccounts/variables', variables('automationAccountName'), 'SentinelTrainingLocation')]",
      "[resourceId('Microsoft.Automation/automationAccounts/variables', variables('automationAccountName'), 'SentinelTrainingWorkspaceName')]"
    ],
    "properties": {
      "runbook": {
        "name": "IngestTrainingTelemetry"
      },
      "schedule": {
        "name": "[parameters('scheduleName')]"
      },
      "parameters": {

```

```

"SubscriptionId": "[subscription().subscriptionId]",

"ResourceGroupName": "[resourceGroup().name]",

"Location": "[parameters('location')]",

"WorkspaceName": "[parameters('WorkspaceName')]"

}

}

    },
    "outputs": {
        "principalId": {
            "type": "string",
            "value": "[reference(resourceId('Microsoft.Automation/automationAccounts', variables('automationAccountName')), '2024-10-23', 'Full').identity.principalId]"
        }
    },
    "parameters": {
        "WorkspaceName": {
            "type": "string",
            "value": "[parameters('workspaceName')]"
        },
        "scheduleStartTime": {
            "type": "string",
            "value": "[parameters('scheduleStartTime')]"
        },
        "scheduleName": {
            "type": "string",
            "value": "[variables('scheduleName')]"
        },
        "location": {
            "type": "string",
            "value": "[parameters('workspace-location')]"
        }
    },
    "type": "Microsoft.Authorization/roleAssignments",
    "apiVersion": "2022-04-01",
    "name": "[guid(resourceGroup().id, variables('automationAccountName'), deployment().name, 'monitoring-contrib')]",
    "condition": "[parameters('enableIngestion')]",
    "dependsOn": [
        "[resourceId('Microsoft.Resources/deployments', 'ingestEvents')]"
    ]
}

```

```

        },
        "properties": {
            "roleDefinitionId": "[subscriptionResourceId('Microsoft.Authorization/roleDefinitions', '749f88d5-cbae-40b8-bcfc-e573ddc772fa')]",
            "principalId": "[reference('ingestEvents').outputs.principalId.value]",
            "principalType": "ServicePrincipal"
        }
    },
    {
        "type": "Microsoft.Authorization/roleAssignments",
        "apiVersion": "2022-04-01",
        "name": "[guid(resourceGroup().id, variables('automationAccountName'), deployment().name, 'la-contrib')]",
        "condition": "[parameters('enableIngestion')]",
        "dependsOn": [
            "[resourceId('Microsoft.Resources/deployments', 'ingestEvents')]"
        ],
        "properties": {
            "roleDefinitionId": "[subscriptionResourceId('Microsoft.Authorization/roleDefinitions', '92aaf0da-9dab-42b6-94a3-d43ce8d16293')]",
            "principalId": "[reference('ingestEvents').outputs.principalId.value]",
            "principalType": "ServicePrincipal"
        }
    },
    {
        "type": "Microsoft.Authorization/roleAssignments",
        "apiVersion": "2022-04-01",
        "name": "[guid(resourceGroup().id, variables('automationAccountName'), deployment().name, 'metrics-publisher')]",
        "condition": "[parameters('enableIngestion')]",
        "dependsOn": [
            "[resourceId('Microsoft.Resources/deployments', 'ingestEvents')]"
        ],
        "properties": {
            "roleDefinitionId": "[subscriptionResourceId('Microsoft.Authorization/roleDefinitions', '3913510d-42f4-4e42-8a64-420c390055eb')]",
            "principalId": "[reference('ingestEvents').outputs.principalId.value]",
            "principalType": "ServicePrincipal"
        }
    },
    {
        "name": "alertRules",
        "type": "Microsoft.Resources/deployments",
        "apiVersion": "2024-07-01",
        "dependsOn": [
            "[resourceId('Microsoft.Resources/deployments', 'workbook')]",
            "[resourceId('Microsoft.Resources/deploymentScripts', 'waitForSentinelOnboarding')]"
        ]
    }
]

```

```

    },
    "properties": {
      "mode": "Incremental",
      "templateLink": {
        "uri": "[uri(variables('_a
rtifactsLocation'), concat('Artifacts/LinkedTemplates/alertRules.json?v=', parameters('artifa
ctsVersion')))]",
        "contentVersion": "1.0.0.
0"
      }
    },
    "parameters": {
      "workspaceName": {
        "valu
e": "[parameters('workspaceName')]"
      }
    }
  },
  {
    "name": "playbook",
    "type": "Microsoft.Resources/deployments",
    "apiVersion": "2024-07-01",
    "dependsOn": [
      "[resourceId('Microsoft.Resources/deployment
s','workspaceCreation')]"
    ],
    "properties": {
      "mode": "Incremental",
      "templateLink": {
        "uri": "[uri(variables('_a
rtifactsLocation'), concat('Artifacts/LinkedTemplates/playbook.json?v=', parameters('artifact
sVersion')))]",
        "contentVersion": "1.0.0.
0"
      }
    },
    "parameters": {
      "PlaybookName": {
        "value":
"[parameters('playbook1-PlaybookName')]"
      }
    }
  },
  {
    "name": "workbook",
    "type": "Microsoft.Resources/deployments",
    "apiVersion": "2024-07-01",
    "dependsOn": [
      "[resourceId('Microsoft.Resources/deployments','i
ngestEvents')]"
    ],
    "properties": {
      "mode": "Incremental",
      "templateLink": {

```

```

        "uri": "[uri(variables('_a
rtifactsLocation'), concat('Artifacts/LinkedTemplates/workbook.json?v=', parameters('artifact
sVersion')))]",
        "contentVersion": "1.0.0.
0"
    },
    "parameters": {
        "workbookDisplayName": {
            "value": "[concat(parameters('workbook1-name'), ' - ', parameters('formattedTimeNow'))]"
        },
        "workbookSourceId": {
            "value": "[reference('workspaceCreation').outputs.workspaceResourceIdOutput.value]"
        }
    }
},
{
    "name": "workbookLakeUsage",
    "type": "Microsoft.Resources/deployments",
    "apiVersion": "2024-07-01",
    "dependsOn": [
        "[resourceId('Microsoft.Resources/deployments', 'i
ngestEvents')]"
    ],
    "properties": {
        "mode": "Incremental",
        "templateLink": {
            "uri": "[uri(variables('_a
rtifactsLocation'), concat('Artifacts/LinkedTemplates/WorkspaceLakeUsage-ARM.json?v=', parame
ters('artifactsVersion')))]",
            "contentVersion": "1.0.0.
0"
        },
        "parameters": {
            "workbookDisplayName": {
                "value": "[concat('Workspace & Lake Usage - ', parameters('formattedTimeNow'))]"
            },
            "WorkspaceId": {
                "value": "[reference('workspaceCreation').outputs.workspaceResourceIdOutput.value]"
            }
        }
    }
},
{
    "name": "watchlist",
    "type": "Microsoft.Resources/deployments",
    "apiVersion": "2024-07-01",
    "dependsOn": [
        "[resourceId('Microsoft.Resources/deploymentScri
pts', 'waitForSentinelOnboarding')]"
    ]
}

```

```

    },
    "properties": {
      "mode": "Incremental",
      "templateLink": {
        "uri": "[uri(variables('_a
rtifactsLocation'), concat('Artifacts/LinkedTemplates/watchlist.json?v=', parameters('artifac
tsVersion')))]",
        "contentVersion": "1.0.0.
0"
      },
      "parameters": {
        "workspaceName": {
          "valu
e": "[parameters('workspaceName')]"
        }
      }
    },
    "outputs": {
    }
  }
}

```

4. Once complete, go back to Microsoft Sentinel — you should see ingested data and recent incidents on the home page.

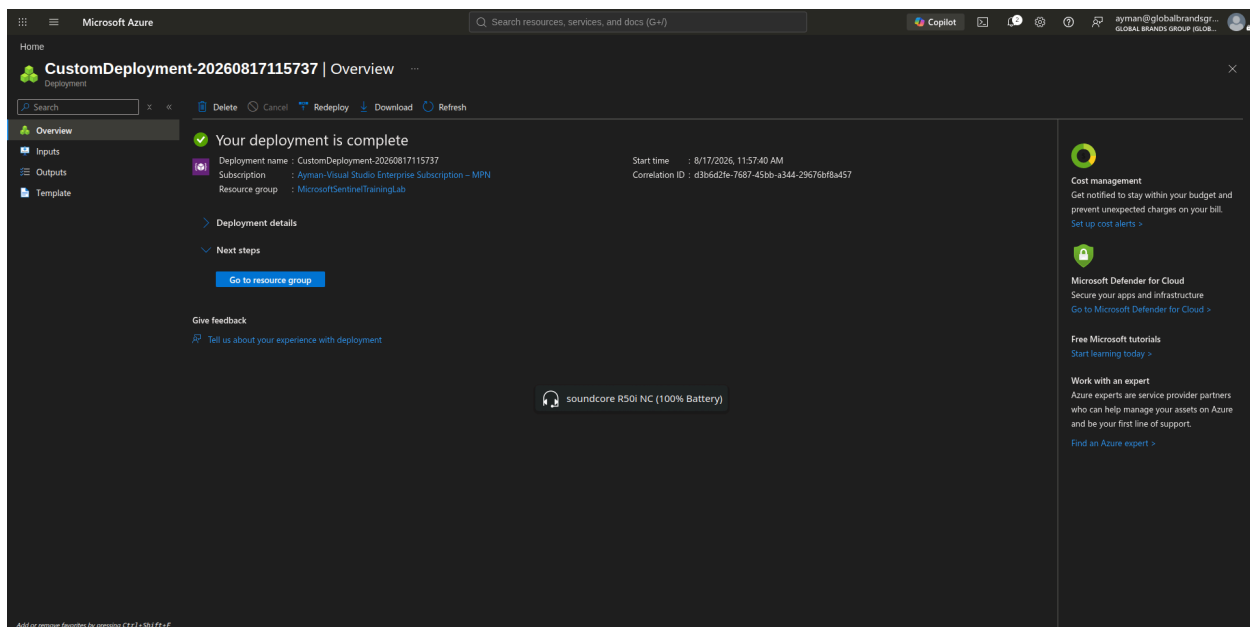
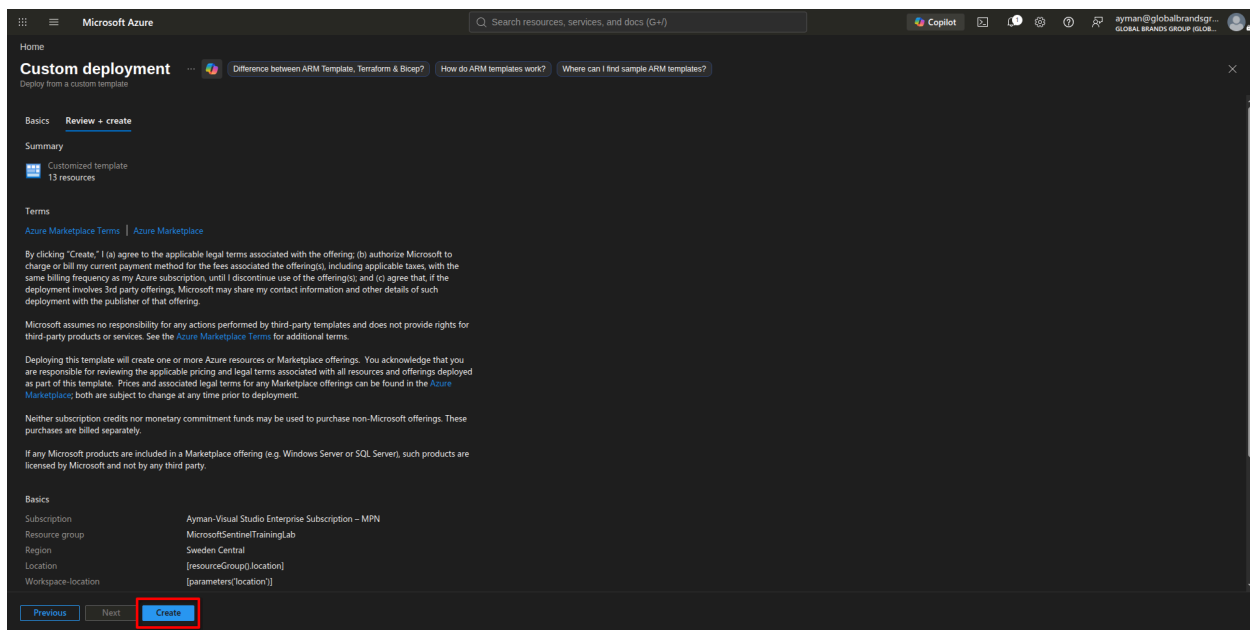
⚠️ Playbook Permissions: The deployment creates a Logic App with a **System-Assigned Managed Identity**. Grant this identity the **Microsoft Sentinel Contributor** role on the resource group for the playbook to run automatically:

1. **Resource Group** → **Access control (IAM)** → **Add role assignment**.
2. Select **Microsoft Sentinel Automation Contributor**, assign to **Managed identity**, select the Logic App's identity, and **Save**.

The screenshot shows the 'Custom deployment' wizard in the Microsoft Azure portal. The 'Edit template' button is highlighted with a red box. The form includes the following fields:

- Subscription:** Ayman-Visual Studio Enterprise Subscription - MPN
- Resource group:** (empty, with a 'Create new' link)
- Region:** (Europe) Sweden Central
- Location:** [resourceGroup().location]
- Workspace location:** [parameters('location')]
- Workspace Name:** (empty)
- Formatted Time Now:** [utcNow('g')]
- Artifacts Version:** [utcNow('yyyyMMddHHmmss')]
- Schedule Start Time:** [dateTimeAdd(utcNow('yyyy-MM-ddTHH:mm:ssZ'), 'PT15M')]
- Detection Rules Schedule Start Time:** [dateTimeAdd(utcNow('yyyy-MM-ddTHH:mm:ssZ'), 'PT30M')]

At the bottom, there are buttons for 'Previous', 'Next', and 'Review + create'.



Home > MicrosoftSentinelTrainingLab | Access control (IAM) ☆

Search [] + Add Download role assignments Edit columns Refresh Delete Feedback

Overview Activity log **Access control (IAM)** Tags Resource visualizer Events Settings Cost Management Cost analysis Cost alerts (preview) Budgets Advisor recommendations Monitoring Insights (preview) Alerts Metrics Diagnostic settings Logs Advisor recommendations Workbooks Automation Help

Add role assignment

user has elevated access in your tenant. You should take immediate action and remove all role assignments with elevated access. In addition, remove any role assignments at root scope that are no longer needed. [View role assignments](#)

Check access Role assignments Roles Deny assignments

Looking for the previous check access view? [Click here](#)

Check access
Review the level of access a user, group, service principal, or managed identity has to this resource. [Learn more](#)

[Check access](#)

ayman@globalbrandsgroup448.onmicrosoft.com Assignments

Active 2 Eligible 0 Deny 0

Search by role name or membership

Role Name	Scope	Membership	Condition	Action
Active permanent assignments (2)				
Foundry User	Subscription (Inherited)	ayman	No	-
Owner	Subscription (Inherited)	ayman	Yes	-
Active time-bound assignments (0)				
Eligible permanent assignments (0)				
Eligible time-bound assignments (0)				
Deny assignments (0)				

Add or remove favorites by pressing Ctrl+Shift+F

Microsoft Azure Search resources, services, and docs (G+I) Copilot ayman@globalbrandsgr... GLOBAL BRANDS GROUP (GLOB...

Home > MicrosoftSentinelTrainingLab | Access control (IAM)

Add role assignment

Role Members Conditions Assignment type Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

[Copilot can help pick a role](#)

Job function roles Privileged administrator roles

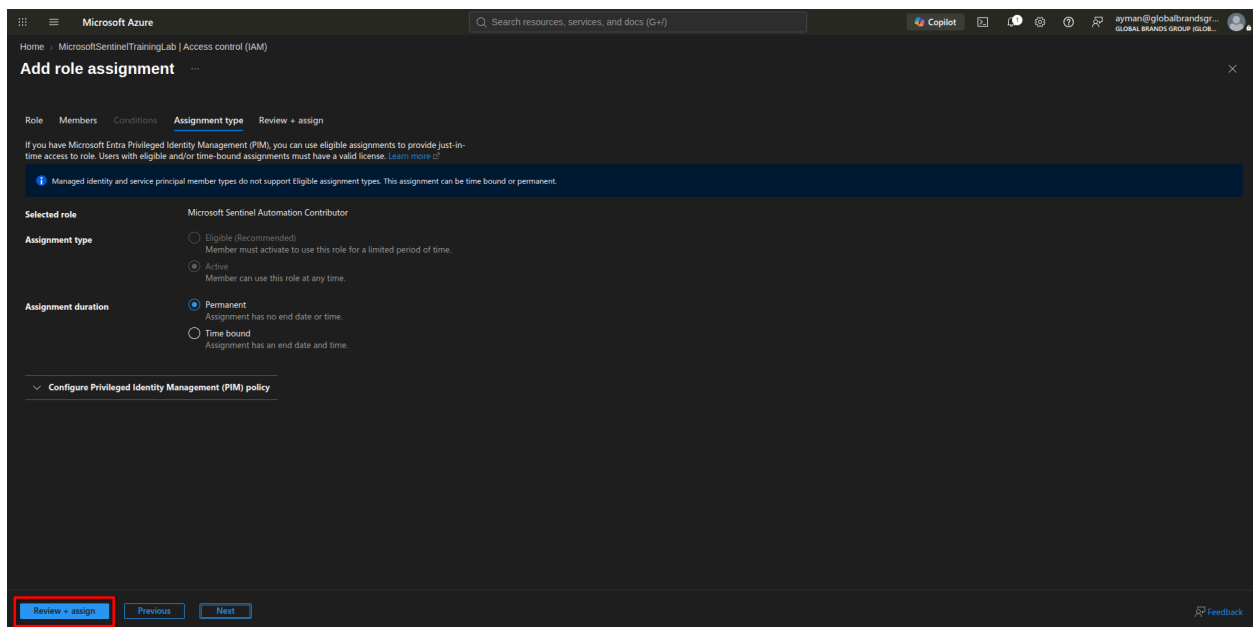
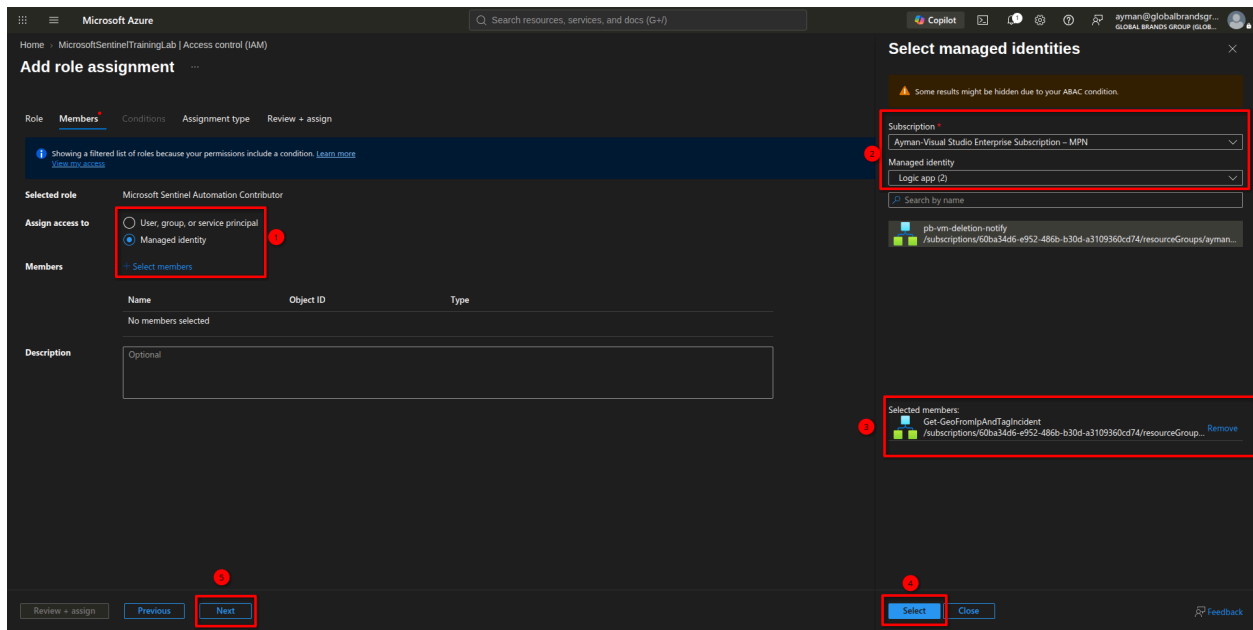
Grant access to Azure resources based on job function, such as the ability to create virtual machines.

Microsoft Sentinel Contributor × Type: All Category: All

Name	Description	Type	Category	Details
Microsoft Sentinel Automation Contributor	Microsoft Sentinel Automation Contributor	BuiltInRole	Security	View
Microsoft Sentinel Contributor	Microsoft Sentinel Contributor	BuiltInRole	Security	View

Showing 1 - 2 of 2 results.

Review + assign Previous **Next** Feedback



Exploration: Hunting Across Your Data

Discover Your Tables

Open the **Microsoft Defender portal** → **Hunting** → **Advanced Hunting**.

Run the following query to see which tables contain data and how much:

```
search *
| summarize EventCount = count() by $table
| sort by EventCount desc
```

The screenshot shows the Microsoft Defender Advanced Hunting interface. The left sidebar contains navigation options like Home, Cases, Exposure management, Investigation & response, Threat intelligence, Assets, Microsoft Sentinel, Identities, Endpoints, Email & collaboration, Cloud apps, Cloud security, SOC optimization, Reports, Learning hub, Trials, More resources, and System. The main panel is titled 'Advanced hunting' and shows a query editor with the following query:

```
1 search *
2 | summarize EventCount = count() by $table
3 | sort by EventCount desc
```

The query results are displayed in a table with 38 items. The table has columns for Name, SeverityName, Tactic, and AlertCount. The results are sorted by AlertCount in descending order.

Name	SeverityName	Tactic	AlertCount
ExposureGraphNo...	1022577		
DeviceTvmSoftwar...	328619		
SecurityEvent	23864		
ExposureGraphEd...	4348		
DeviceTvmSoftwar...	2508		
GraphAPIAuditEve...	531		
AzureActivity_CL	325		
OfficeActivity_CL	266		
CommonSecurityL...	149		
DeviceTvmSecure...	117		
EntraIDSignInEv...	108		
DeviceTvmInfoGat...	92		
CloudAppEvents	93		

Explore CrowdStrike Endpoint Data

CrowdStrike is the EDR (Endpoint Detection and Response) solution in this lab. Let's explore the alerts it generated:

```
CrowdStrikeAlerts
```

```
| summarize AlertCount = count() by Name, SeverityName, Tactic
| sort by AlertCount desc
```

This shows you the types of alerts CrowdStrike raised. Notice the variety of **MITRE ATT&CK tactics** — from Initial Access through Exfiltration — indicating a multi-stage attack.

The screenshot shows the Microsoft Defender Advanced Hunting interface. The left sidebar contains navigation options like Home, Cases, Exposure management, Investigation & response, Threat intelligence, Assets, Microsoft Sentinel, Identities, Endpoints, Email & collaboration, Cloud apps, Cloud security, SOC optimization, Reports, Learning hub, Trials, More resources, and System. The main panel is titled 'Advanced hunting' and shows a query editor with the following query:

```
1 CrowdStrikeAlerts
2 | summarize AlertCount = count() by Name, SeverityName, Tactic
3 | sort by AlertCount desc
```

The query results are displayed in a table with 9 items. The table has columns for Name, SeverityName, Tactic, and AlertCount. The results are sorted by AlertCount in descending order.

Name	SeverityName	Tactic	AlertCount
Malicious PowerSh...	Critical	Execution	12
Credential Dumpi...	Critical	Credential Access	10
Sensitive File Stag...	Medium	Collection	9
Security Tool Tamp...	High	Defense Evasion	6
C2 Beacon Detected	Critical	Command and Control	6
Ransomware Beha...	Critical	Impact	5
Lateral Movement ...	High	Lateral Movement	3
User Execution: M...	Critical	Execution	1
NetworkConnect C2	Critical	Command and Control	1

Now look at the timeline of alerts per device:

```

CrowdStrikeAlerts
| extend DeviceName = tostring(split(DisplayName, " on ")[-1])
| summarize
    AlertCount = count(),
    Tactics = make_set(Tactic),
    FirstAlert = min(TimeGenerated),
    LastAlert = max(TimeGenerated)
    by DeviceName
| sort by AlertCount desc

```

Key insight: If a single device has alerts spanning multiple MITRE tactics (e.g., Execution, Credential Access, Lateral Movement), it is likely compromised and being used as a pivot point in a multi-stage attack.

The screenshot shows the Microsoft Defender Advanced Hunting console. On the left is a navigation pane with categories like Home, Incidents, Advanced hunting, and Investigation & response. The main area is titled 'Advanced hunting' and shows a query editor with the following KQL query:

```

1 CrowdStrikeAlerts
2 | extend DeviceName = tostring(split(DisplayName, " on ")[-1])
3 | summarize
4     AlertCount = count(),
5     Tactics = make_set(Tactic),
6     FirstAlert = min(TimeGenerated),
7     LastAlert = max(TimeGenerated)
8     by DeviceName
9 | sort by AlertCount desc

```

Below the query editor, the 'Results' tab is active, displaying a table with 8 items. The table columns are DeviceName, AlertCount, Tactics, FirstAlert, and LastAlert. The data shows various devices with their respective alert counts and associated MITRE tactics.

DeviceName	AlertCount	Tactics	FirstAlert	LastAlert
win11a	28	["Execution", "Credential ...	Aug 17, 2026 12:24:59 PM	Aug 17, 2026 12:24:59 PM
it-wd01	8	["Execution", "Collection"...	Aug 17, 2026 12:24:59 PM	Aug 17, 2026 12:24:59 PM
srv-9w01	4	["Credential Access", "Im...	Aug 17, 2026 12:24:59 PM	Aug 17, 2026 12:24:59 PM
dev-wd01	4	["Execution", "Defense Ev...	Aug 17, 2026 12:24:59 PM	Aug 17, 2026 12:24:59 PM
win11d	3	["Lateral Movement", "Ex...	Aug 17, 2026 12:24:59 PM	Aug 17, 2026 12:24:59 PM
win11b	3	["Collection", "Execution"...	Aug 17, 2026 12:24:59 PM	Aug 17, 2026 12:24:59 PM
win11c	2	["Execution"]	Aug 17, 2026 12:24:59 PM	Aug 17, 2026 12:24:59 PM
srv-dc01	1	["Impact"]	Aug 17, 2026 12:24:59 PM	Aug 17, 2026 12:24:59 PM

Explore Firewall Traffic (Palo Alto)

The `CommonSecurityLog` table contains Palo Alto firewall logs. Let's get a traffic overview:

```

CommonSecurityLog
| where DeviceVendor == "Palo Alto Networks"
| summarize
    TotalEvents = count(),
    DistinctSources = dcount(SourceIP),
    DistinctDestinations = dcount(DestinationIP)
    by Activity
| sort by TotalEvents desc

```

The screenshot shows the Microsoft Defender Advanced Hunting interface. The left sidebar contains navigation options like Home, Incidents, Investigation & response, Threat intelligence, Assets, Microsoft Sentinel, Identities, Endpoints, Email & collaboration, Cloud apps, Cloud security, SOC optimization, Reports, Learning hub, Trials, More resources, and System. The main panel is titled 'Advanced hunting' and shows a query editor with the following query:

```
1 CommonSecurityLog
2 | where DeviceVendor == "Palo Alto Networks"
3 | summarize
4 |   TotalEvents = count(),
5 |   DistinctSources = dcount(SourceIP),
6 |   DistinctDestinations = dcount(DestinationIP)
7 | by Activity
8 | sort by TotalEvents desc
```

The results table shows 3 items:

Activity	TotalEvents	DistinctSources	DistinctDestinations
drop	125	1	84
end	23	1	4
spyware	1	1	1

Now look for denied traffic — potential reconnaissance or blocked attacks:

```
CommonSecurityLog
| where DeviceVendor == "Palo Alto Networks"
| where Activity in ("drop", "deny", "reset-both")
| summarize
|   BlockedConnections = count(),
|   TargetedPorts = dcount(DestinationPort)
| by SourceIP
| sort by BlockedConnections desc
| take 10
```

The screenshot shows the Microsoft Defender Advanced Hunting interface with a new query:

```
1 CommonSecurityLog
2 | where DeviceVendor == "Palo Alto Networks"
3 | where Activity in ("drop", "deny", "reset-both")
4 | summarize
5 |   BlockedConnections = count(),
6 |   TargetedPorts = dcount(DestinationPort)
7 | by SourceIP
8 | sort by BlockedConnections desc
9 | take 10
```

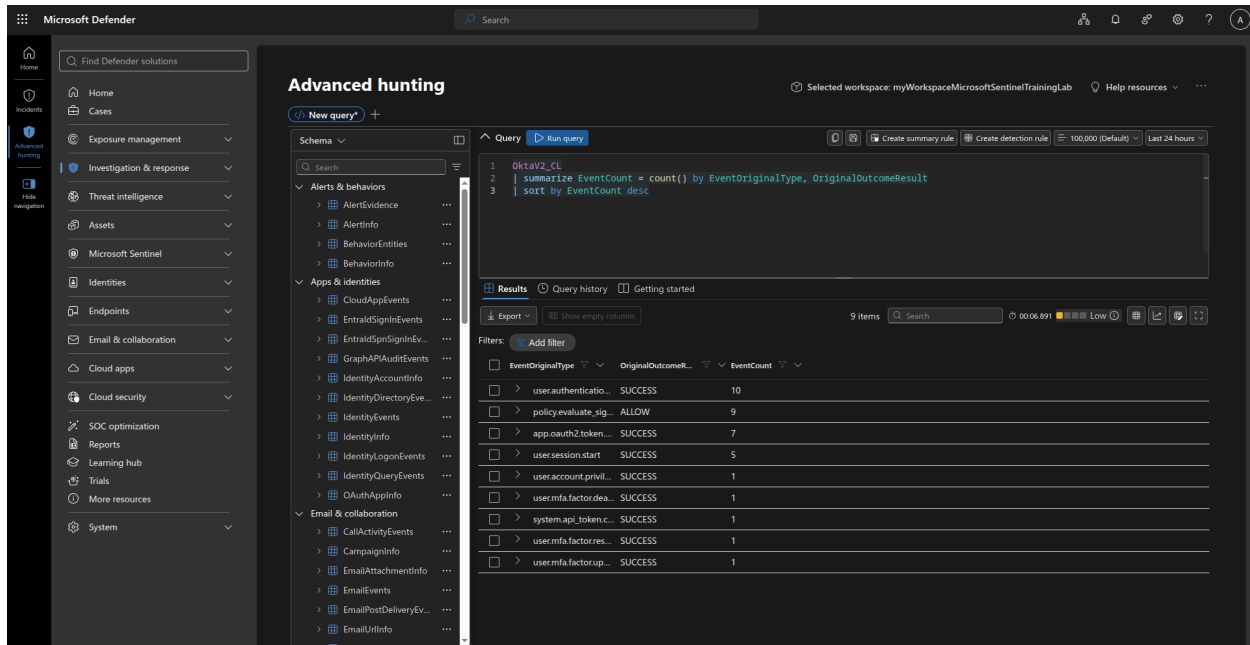
The results table shows 1 item:

SourceIP	BlockedConnections	TargetedPorts
10.0.1.50	125	124

Explore Identity Events (Okta)

The `OktaV2_CL` table contains identity events from Okta. Let's check for suspicious activity:

```
OktaV2_CL
| summarize EventCount = count() by EventOriginalType, OriginalOutcomeResult
| sort by EventCount desc
```



The screenshot shows the Microsoft Defender Advanced Hunting interface. On the left is a navigation pane with categories like Home, Incidents, Advanced hunting, and Data navigation. The 'Advanced hunting' section is expanded, showing various data sources. The main area is titled 'Advanced hunting' and contains a query editor. The query is:

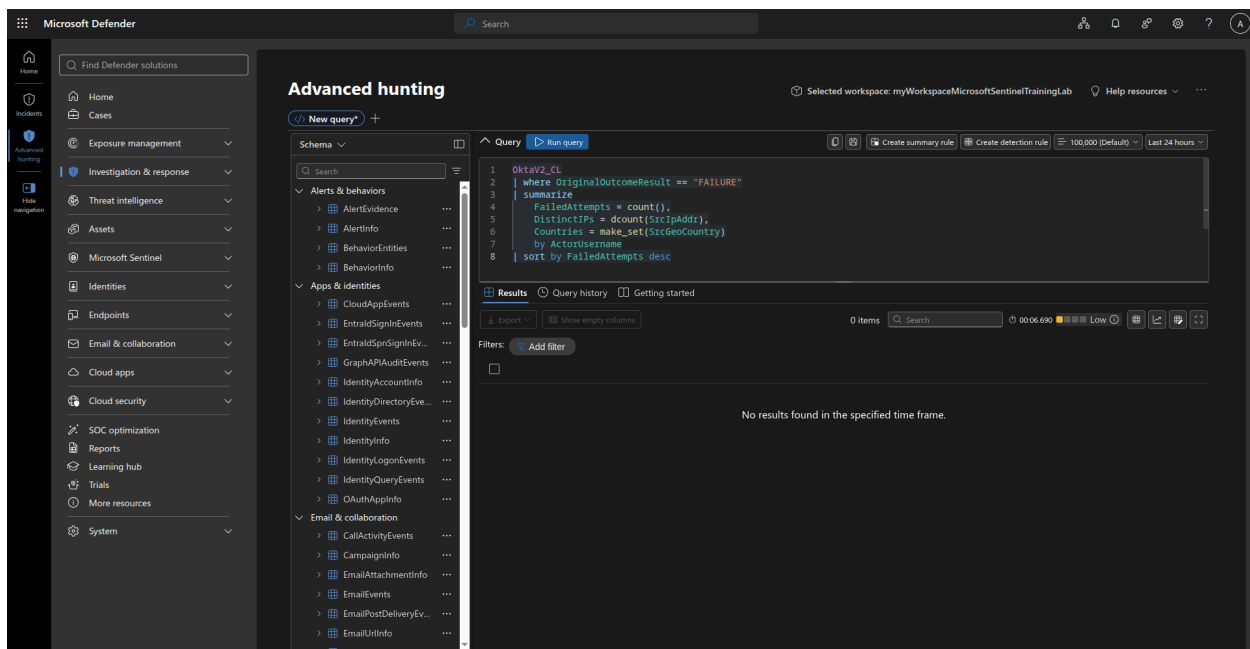
```
1 OktaV2_CL
2 | summarize EventCount = count() by EventOriginalType, OriginalOutcomeResult
3 | sort by EventCount desc
```

Below the query editor, there are tabs for 'Results', 'Query history', and 'Getting started'. The 'Results' tab is active, showing a table with 9 items. The table has columns for 'EventOriginalType', 'OriginalOutcomeResult', and 'EventCount'. The data is as follows:

EventOriginalType	OriginalOutcomeResult	EventCount
user.authentication...	SUCCESS	10
policy.evaluate_sig...	ALLOW	9
app.oauth2.token...	SUCCESS	7
user.session.start	SUCCESS	5
user.account.privat...	SUCCESS	1
user.mfa.factor.dea...	SUCCESS	1
system.api.token.c...	SUCCESS	1
user.mfa.factor.res...	SUCCESS	1
user.mfa.factor.up...	SUCCESS	1

Look for failed logins and MFA-related events:

```
OktaV2_CL
| where OriginalOutcomeResult == "FAILURE"
| summarize
    FailedAttempts = count(),
    DistinctIPs = dcount(SrcIpAddr),
    Countries = make_set(SrcGeoCountry)
    by ActorUsername
| sort by FailedAttempts desc
```



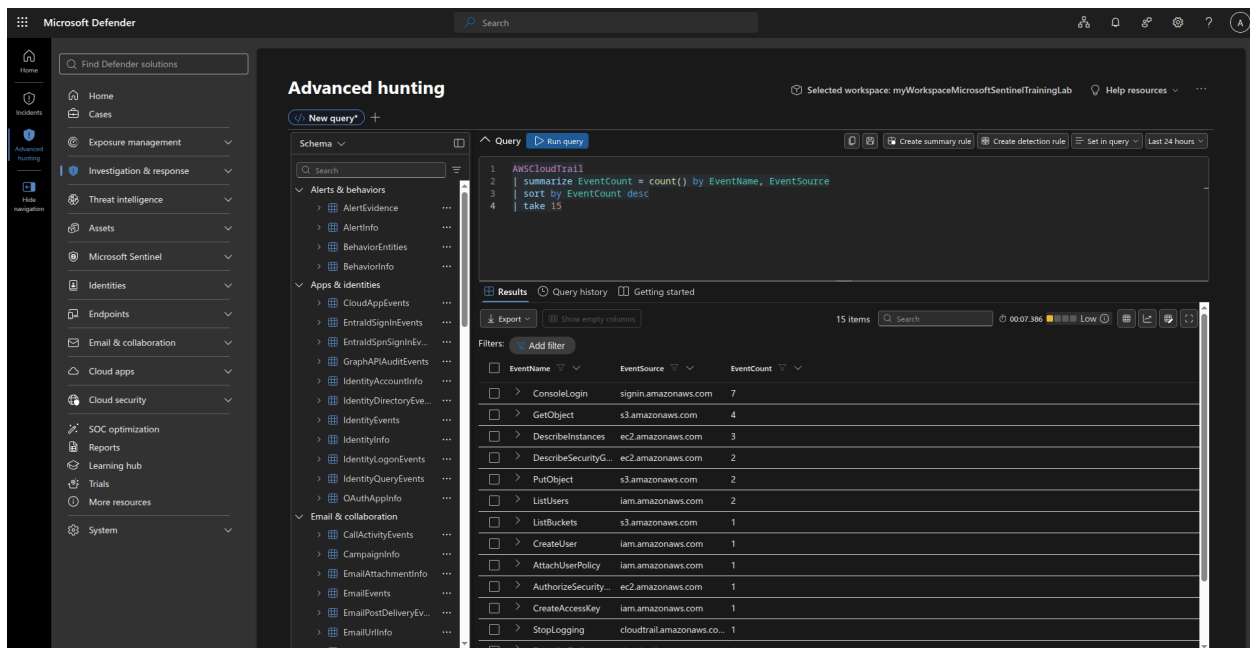
Explore Cloud Activity (AWS)

The `AWSCloudTrail` table tracks API calls in AWS. Check what happened:

```

AWSCloudTrail
| summarize EventCount = count() by EventName, EventSource
| sort by EventCount desc
| take 15

```

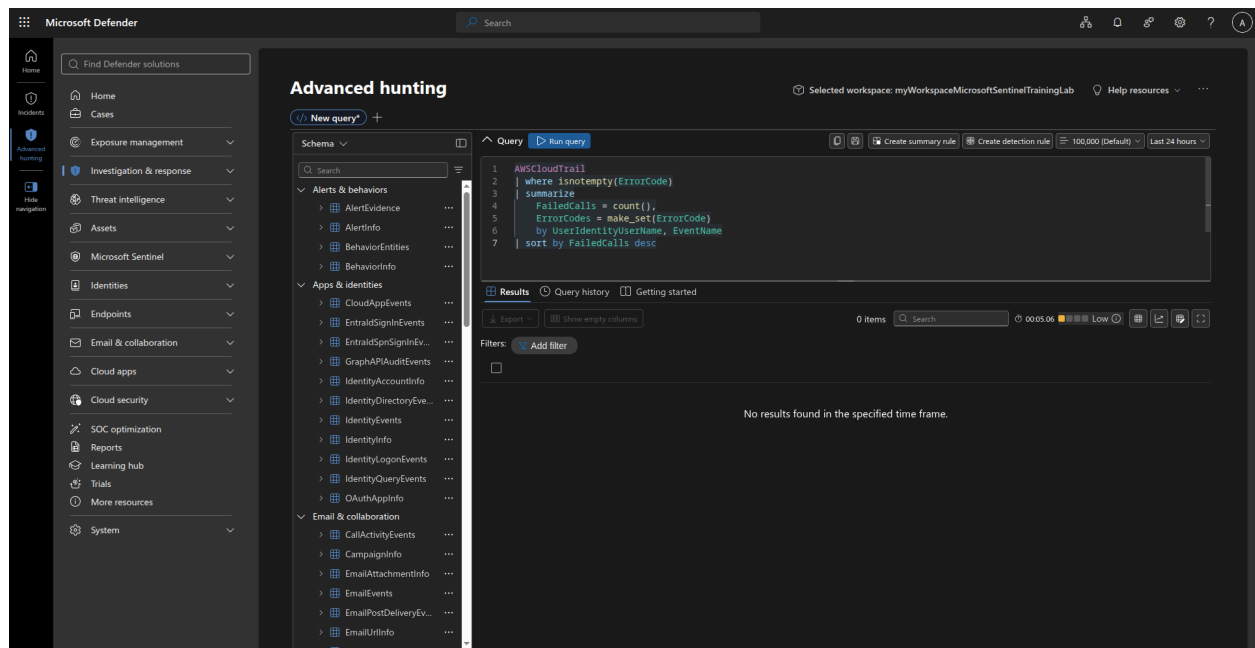


Look for failed operations — potential attacker reconnaissance or privilege escalation attempts:


```

AWSCloudTrail
| where isnotempty(ErrorCode)
| summarize
    FailedCalls = count(),
    ErrorCodes = make_set(ErrorCode)
    by UserIdentityUserName, EventName
| sort by FailedCalls desc

```



Build a Cross-Source Summary

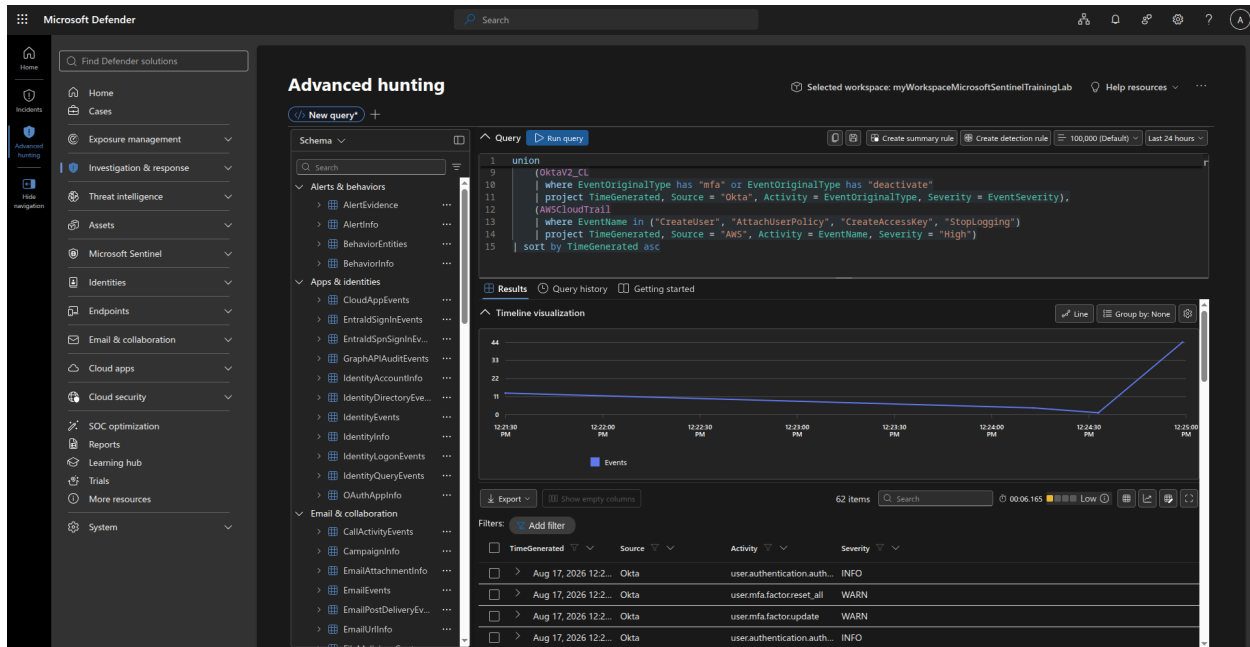
Now combine insights from multiple sources to understand the full attack scope:

```

union
(
    CrowdStrikeAlerts
    | where SeverityName in ("Critical", "High")
    | project TimeGenerated, Source = "CrowdStrike", Activity = Name, Severity = SeverityName
),
(
    CommonSecurityLog
    | where DeviceVendor == "Palo Alto Networks"
    | where DeviceEventClassID == "THREAT"
    | project TimeGenerated, Source = "Palo Alto", Activity = Activity, Severity = LogSeverity
),
(
    OktaV2_CL
    | where EventOriginalType has "mfa" or EventOriginalType has "deactivate"
    | project TimeGenerated, Source = "Okta", Activity = EventOriginalType, Severity = EventSeverity
),
(
    AWSCloudTrail
    | where EventName in ("CreateUser", "AttachUserPolicy", "CreateAccessKey", "StopLogging")
    | project TimeGenerated, Source = "AWS", Activity = EventName, Severity = "High"
)
| sort by TimeGenerated asc

```

This gives you a **chronological timeline** of suspicious activity across all data sources — the foundation of any investigation.



Create a Custom Detection Rule

Now that you understand the data, let's turn a hunting query into a detection rule. You'll create a rule that detects **multi-tactic activity on a single endpoint** — a strong indicator of a compromised device.

Write the Detection Query

This query finds devices with CrowdStrike alerts spanning 3 or more MITRE ATT&CK tactics within a 4-hour window:

```
CrowdStrikeAlerts
| where TimeGenerated > ago(4h)
| where SeverityName in ("Critical", "High")
| extend DeviceName = tostring(split(DisplayName, " on ")[-1])
| summarize
    TacticCount = dcount(Tactic),
    Tactics = make_set(Tactic),
    AlertNames = make_set(Name, 10),
    AlertCount = count(),
    FirstSeen = min(TimeGenerated),
    LastSeen = max(TimeGenerated)
    by DeviceName, AgentId
| where TacticCount >= 3
| project
    TimeGenerated = FirstSeen,
    DeviceName,
    TacticCount,
    Tactics,
    AlertNames,
    AlertCount,
    FirstSeen,
    LastSeen,
    ReportId = tostring(hash_sha256(strcat(DeviceName, tostring(FirstSeen))))
```

Run this query in Advanced Hunting to verify it returns results.

The screenshot shows the Microsoft Defender Advanced Hunting interface. On the left is a navigation pane with categories like Home, Incidents, Investigation & response, Threat intelligence, Assets, Microsoft Sentinel, Identities, Endpoints, Email & collaboration, Cloud apps, Cloud security, SOC optimization, Reports, Learning hub, Trials, More resources, and System. The main area is titled 'Advanced hunting' and shows a 'New query' button. A query is entered in the 'Query' field:

```
1 CrowdStrikeAlerts
2 | where TimeGenerated > ago(4h)
3 | where SeverityName in ("Critical", "High")
4 | extend DeviceName = tostring(split(DisplayName, " on ")[-1])
5 | summarize
6 |   TacticCount = dcount(Tactics),
7 |   Tactics = make_set(Tactics),
8 |   AlertNames = make_set(Name, 10),
9 |   AlertCount = count()
10 | by DeviceName, make_set(DeviceName, 10)
```

Below the query, the 'Results' tab shows 4 items. The table has columns: TimeGenerated, DeviceName, TacticCount, Tactics, AlertNames, AlertCount, and FirstSeen.

TimeGenerated	DeviceName	TacticCount	Tactics	AlertNames	AlertCount	FirstSeen
Aug 17, 2026 12:2...	win11a	5	["Execution","Credential ...	["User Execution: Malici...	24	Aug 17, 2...
Aug 17, 2026 12:2...	srv-file01	3	["Credential Access","Im...	["Credential Dumping vi...	4	Aug 17, 2...
Aug 17, 2026 12:2...	dev-ws01	3	["Execution","Defense Ev...	["Malicious PowerShell E...	4	Aug 17, 2...
Aug 17, 2026 12:2...	win11d	3	["Lateral Movement","Ex...	["Lateral Movement via ...	3	Aug 17, 2...

Create the Rule

1. After running the query, select **Create detection rule**

This screenshot is identical to the previous one, but the 'Create detection rule' button in the top right of the query editor area is highlighted with a red box.

2. Fill in the rule details:

Field	Value
Name	Lab Stage E1 - Multi-Tactic Compromise on Single Device
Description	Detects when a single endpoint has CrowdStrike alerts spanning 3 or more MITRE ATT&CK tactics, indicating a multi-stage compromise.

Field	Value
Severity	High
MITRE ATT&CK	Leave default (auto-detected)

3. Set the **Schedule**:

- **Frequency** → Every 1 hour
- **Lookback** → 4 hours

3. On the **Impacted entities** tab, map:

- **Device** → `DeviceName`

4. On the **Actions** tab, leave empty for now (no automated response)

5. Select **Create** to save the rule

The screenshot shows the 'Custom detection' configuration page in the Microsoft Defender console. The left sidebar contains navigation links: General (selected), Alert settings, Automated actions, Scope, and Review and create. The main panel is titled 'General' and includes the following sections:

- Detection name:** A text box containing 'Lab Stage E1 - Multi-Tactic Compromise on Single Device'.
- Rule Description:** A text box containing 'Detects when a single endpoint has CrowdStrike alerts spanning 3 or more MITRE ATT&CK tactics, indicating a multi-stage compromise.'
- Rule query:** A text box containing a KQL query:


```

CrowdStrikeAlerts
| where TimeGenerated > ago(4h)
| where SeverityName in ('Critical', 'High')
| extend DeviceName = tostring(split(DisplayName, ' on ')[-1])
| summarize
  TacticsCount = count(Tactics),
  Tactics = make_set(Tactics),
  AlertTimes = make_set(Time, 10),
  AlertCount = count(),
  FirstSeen = min(TimeGenerated),
  LastSeen = max(TimeGenerated)
  by DeviceName, AgentId
| where TacticsCount >= 3
| project
  TimeGenerated = FirstSeen,
  DeviceName,
  TacticsCount,
  Tactics,
  AlertTimes,
  AlertCount,
  FirstSeen,
  LastSeen,
  ReportId = tostring(hash_sha256 strcat(DeviceName, tostring(FirstSeen)))
      
```
- Frequency:** A dropdown menu set to 'Every hour'.

At the bottom of the panel are 'Next' and 'Cancel' buttons. A link 'Create analytics rule instead' is visible in the top right corner of the main panel.

Microsoft Defender

Search

Custom detection

General

Alert settings

Automated actions

Scope

Review and create

Lookback

4

Hours

Severity

High

MITRE ATT&CK

Select one tactic for your rule and the related technique or sub-technique. To change the tactic, deselect it and then select a new one.

Search

Impact

Impair Process Control

Inhibit Response Function

Initial Access

Lateral Movement (15/18 techniques selected)

Persistence

Privilege Escalation

Reconnaissance

Resource Development

Threat analytics report

Select Threat

Recommended actions

Describe suggested investigation or remediation steps

Next

Cancel

Microsoft Defender

Search

Custom detection

General

Alert settings

Automated actions

Scope

Review and create

Alert settings

Define the alert enrichment details.

Explore query and results

Alert details

Alert title

CrowdStrike multi-stage compromise suspected on {{DeviceName}}

Description

The endpoint generated High or Critical CrowdStrike alerts across {{TacticCount}} MITRE ATT&CK tactics in the last 4 hours.

Tactics observed: {{Tactics}}

Custom details

Entity mapping

Identified entities determine how alerts are grouped into incidents. Make sure to carefully review the entities to ensure a high quality of incident correlation. [Read more](#)

Impacted Assets

Entity

Device

Back

Next

Cancel

Microsoft Defender

Search

Custom detection

General

Alert settings

Automated actions

Scope

Review and create

Alert title

CrowdStrike multi-stage compromise suspected on {{DeviceName}}

Description *

The endpoint generated High or Critical CrowdStrike alerts across {{TacticCount}} MITRE ATT&CK tactics in the last 4 hours.
Tactics observed: {{Tactics}}

Custom details

Entity mapping *

Identified entities determine how alerts are grouped into incidents. Make sure to carefully review the entities to ensure a high quality of incident correlation. [Read more](#)

Impacted Assets *

Entity

Device

Identifier *

HostName

Column *

DeviceName

+ Add assets

Related Evidence

+ Add entities

Back

Next

Cancel

Microsoft Defender

Search

Custom detection

General

Alert settings

Automated actions

Scope

Review and create

Automated actions

Define actions to automatically take on affected entities in the generated alerts and incidents.

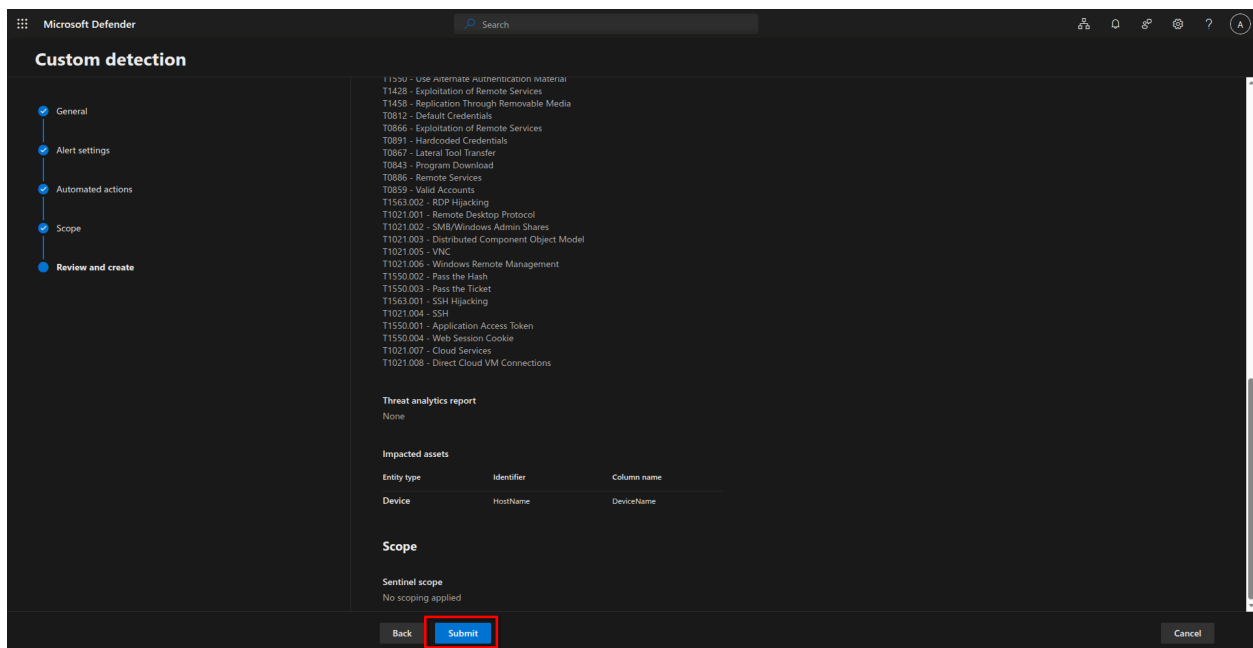
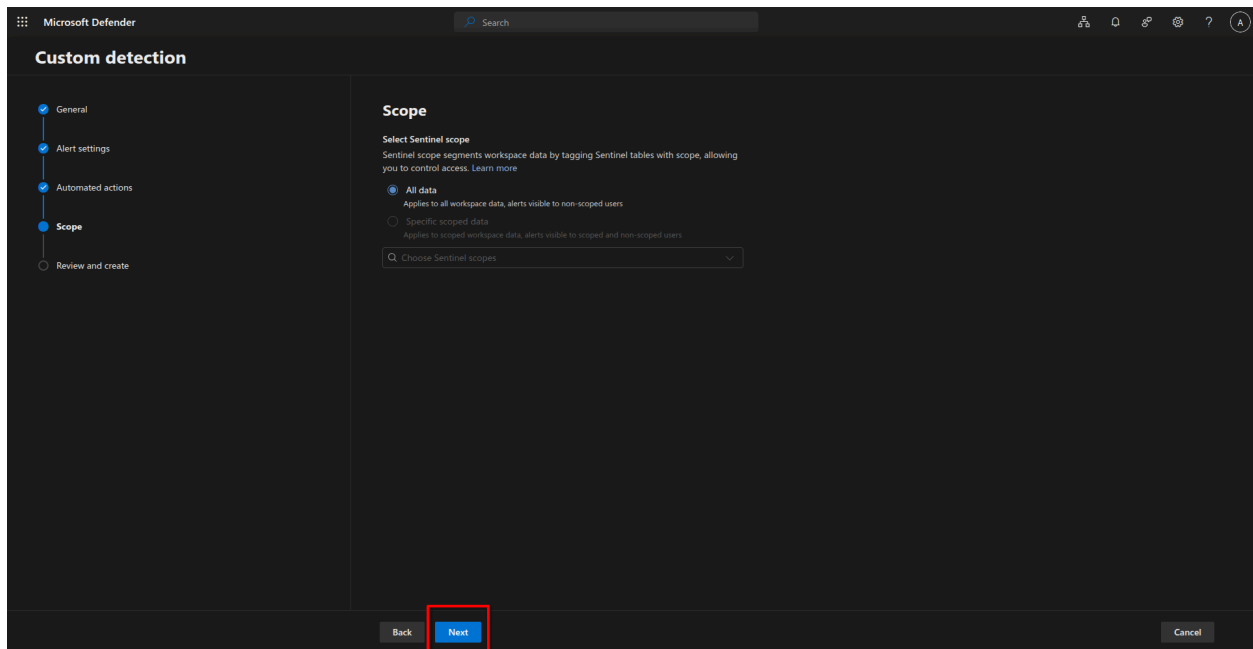
Remediation actions to take

No remediation actions are available for your rule because of missing required columns.

Back

Next

Cancel



Verify the Rule

1. Navigate to **Hunting** → **Custom detection rules**
2. Find your new rule **Lab Stage E1 - Multi-Tactic Compromise on Single Device**
3. Click **Run** to execute it immediately
4. Check **Triggered alerts** to verify the rule fires

Microsoft Defender

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More resources

System

Detection Rules

New rules management experience

Manage your custom detection and analytics rules in one page for a centralized and streamlined rule management experience. This is a new feature that consolidates all your custom rules in one convenient location.

+ Create detection rule ▶ Run ✕ Disable 🗑 Delete ⚙ Disable correlation ⚙ Enable correlation

Filters: Add filter

Name	Severity	Rule type	Incident correlation	Scheduling type
☒ Lab Stage E1 - Multi-Tactic Compr...	High	Custom detection rule	Enabled	Scheduled
☐ AWS Config Service Resource Dele...	Low	Analytics rule	Disabled	Scheduled
☐ Scheduled Task Hide	High	Analytics rule	Disabled	Scheduled
☐ NRT Security Event log cleared	Medium	Analytics rule	Disabled	NRT
☐ Suspicious AWS CLI Command Exe...	Medium	Analytics rule	Disabled	Scheduled
☐ VM Deletion Detected	High	Analytics rule	Disabled	Scheduled
☐ Failedlogon	Medium	Custom detection rule	Enabled	Scheduled
☐ Ibra-Test	Medium	Custom detection rule	Enabled	Scheduled
☐ Malicious DNS Traffic	High	Custom detection rule	Enabled	Scheduled
☐ R	Low	Custom detection rule	Enabled	Scheduled
☐ black cat	High	Custom detection rule	Enabled	Scheduled

Lab Stage E1 - Multi-Tactic Compromise on Single Device

Type: Custom detection rule | High | Enabled

Open detection rule page ▶ **Run** Edit ⚙ Modify query ⋮

General details

Rule Description
Detects when a single endpoint has CrowdStrike alerts spanning 3 or more MITRE ATT&CK tactics, indicating a multi-stage compromise.

Alert Description
The endpoint generated High or Critical CrowdStrike alerts across ([TacticCount]) MITRE ATT&CK tactics in the last 4 hours. Tactics observed: ([Tactics])

Alert title
CrowdStrike multi-stage compromise suspected on ([DeviceName])

Scheduling type
Scheduled

Impacted entities
Device

Provider
Custom

MITRE ATT&CK

Tactic	Technique	Sub-technique
LateralMovement (30)	T0812: Default Credentials	
	T0843: Program Download	
	T0859: Valid Accounts	
	T0866: Exploitation of Remote Services	
	T0867: Lateral Tool Transfer	

Microsoft Defender

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Microsoft Sentinel

Identities

Endpoints

Email & collaboration

Cloud apps

Cloud security

SOC optimization

Reports

Learning hub

Trials

More resources

System

Alerts

Built-in alert tuning was updated with new rules. These rules help streamline your incident and alert queues by prioritizing high-fidelity, actionable alerts. Review more about built-in rules.

Export 1 Week Manage alerts Move alerts

Filter set: Status: New, In progress Add filter Reset all

Alert name	Tags	Severity	Investigation state	Status
☐ NRT Security Event log cleared		Medium		New
☒ CrowdStrike multi-stage compromise suspected o...		High		New
☐ CrowdStrike multi-stage compromise suspected o...		High		New
☐ CrowdStrike multi-stage compromise suspected o...		High		New
☐ CrowdStrike multi-stage compromise suspected o...		High		New
☐ VM Deletion Detected		High		New

CrowdStrike multi-stage compromise suspected on win11d

High | Unknown | New

Open alert page Manage alert ⋮

INSIGHT

Quickly classify this alert

Classify alerts to improve alert accuracy and get more insights about threats to your organization.

Classify alert

Alert state

Classification
Not Set
Set Classification

Assigned to
Unassigned

Alert details

Alert ID
a6c5729032-5e14-4ed3-8670-951feb9c3d0_aad

Categories
Execution, Lateral Movement, Defense Evasion, Persistence, Initial Access

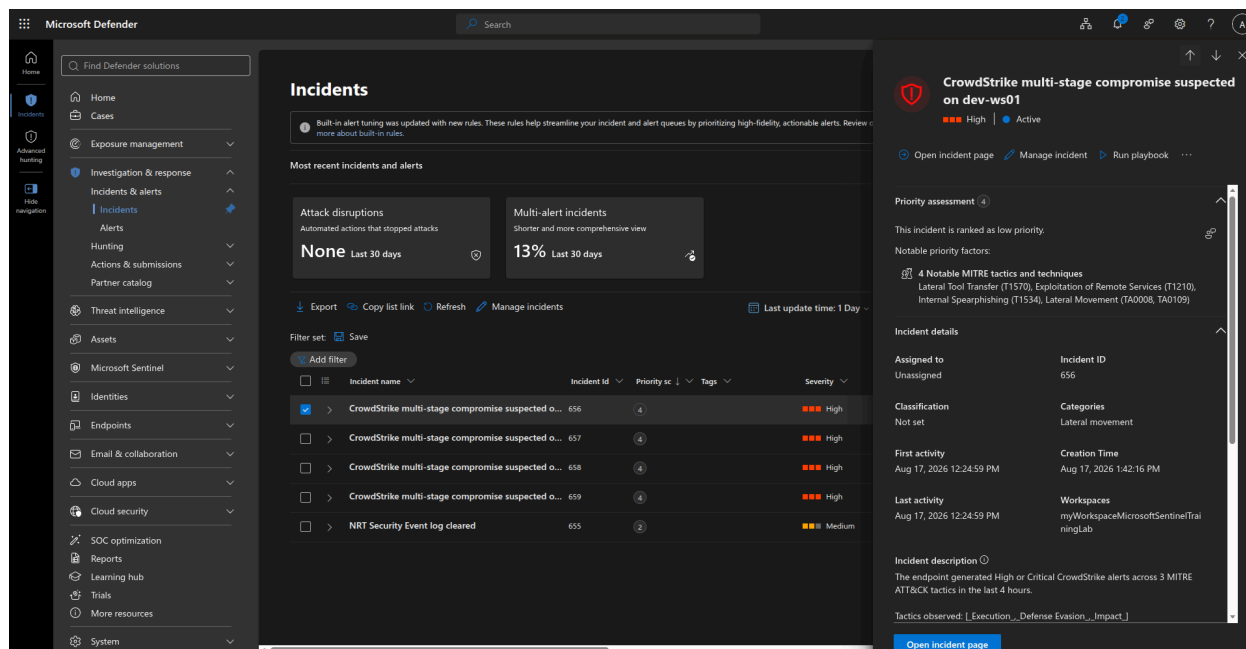
MITRE ATT&CK Techniques
T1210: Exploitation of R... +29 More
View all techniques

Detection source
Custom detection

Service source
Microsoft Defender XDR

Detection technology
-

Open alert page

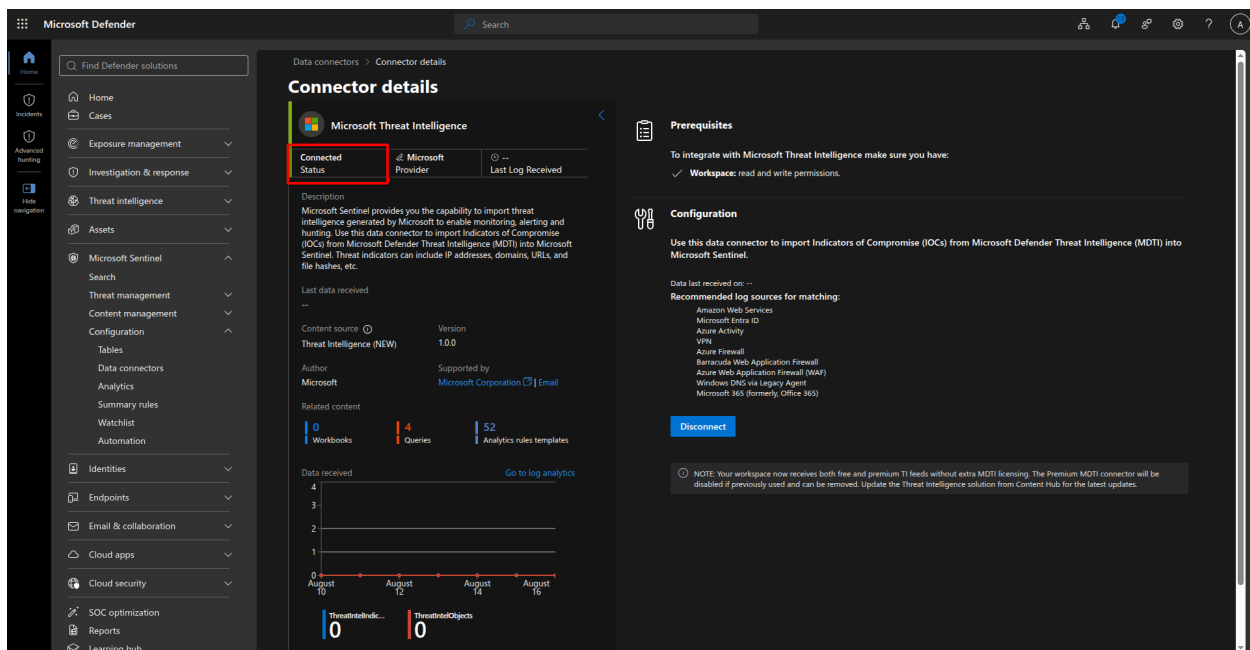
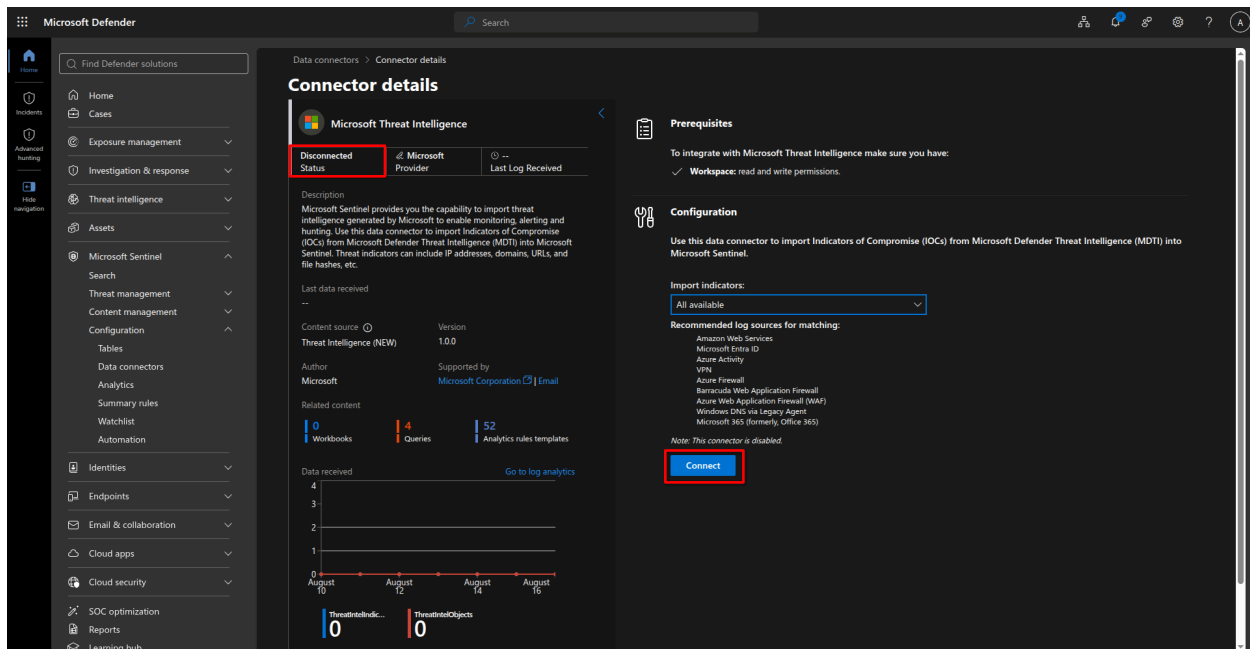


Threat Intelligence: Microsoft Defender Threat Intelligence

Install the Threat Intelligence Solution and Enable the MDTI Data Connector

1. Open the **Microsoft Defender portal** (<https://security.microsoft.com>)
2. Navigate to **Microsoft Sentinel** → **Content management** → **Content hub**
3. Search for **Threat Intelligence** and select the **Threat Intelligence (NEW)** solution (marked as **Featured**)
4. Click **Install/Update** to deploy the solution — this includes the MDTI data connector, analytics rules, workbooks, and hunting queries
5. After installation, navigate to **Microsoft Sentinel** → **Configuration** → **Data connectors**
6. Search for **Microsoft Defender Threat Intelligence**
7. Select the connector and click **Open connector page**
8. Under **Configuration**, select **Connect** to enable the connector
9. Once connected, the **Status** changes to **Connected**. Indicators will begin ingesting within a few minutes.

Note: It can take up to **15 minutes** for the first indicators to appear in the **ThreatIntelIndicators** table. The connector ingests indicators continuously after that.

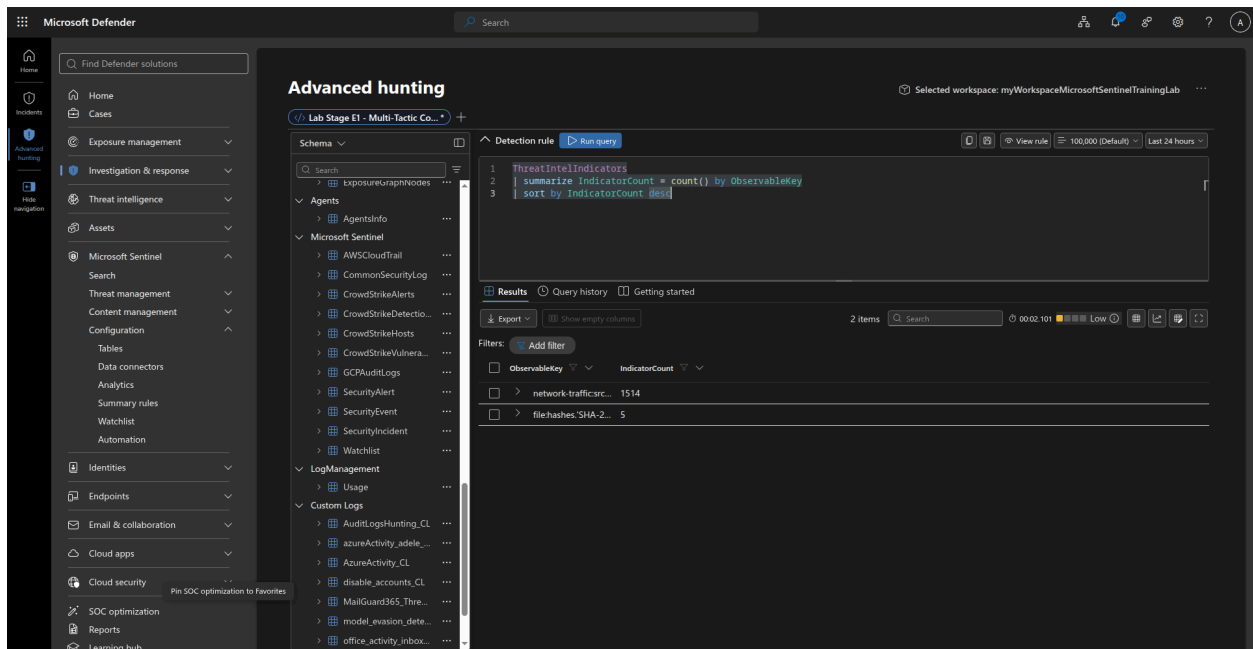


Verify Indicator Ingestion

Once the connector is enabled, verify that indicators are flowing into your workspace:

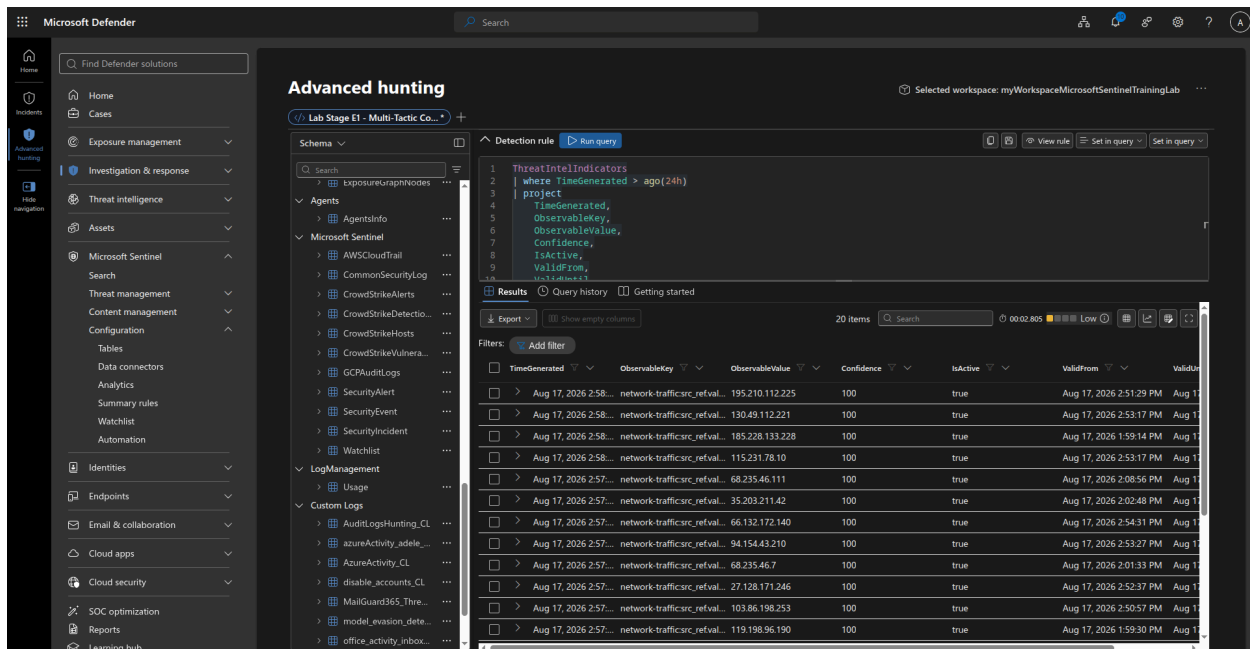
```
ThreatIntelIndicators
| summarize IndicatorCount = count() by ObservableKey
| sort by IndicatorCount desc
```

You should see observable types like `ipv4-addr`, `domain-name`, `url`, and file hash patterns.



Explore the most recent indicators:

```
ThreatIntelIndicators
| where TimeGenerated > ago(24h)
| project
    TimeGenerated,
    ObservableKey,
    ObservableValue,
    Confidence,
    IsActive,
    ValidFrom,
    ValidUntil
| sort by TimeGenerated desc
| take 20
```



Understand the ThreatIntelIndicators Schema

The `ThreatIntelIndicators` table stores STIX-formatted threat intelligence indicators:

Column	Description	Example
<code>ObservableKey</code>	Type of IOC (from the STIX pattern)	<code>ipv4-addr</code> , <code>domain-name</code> , <code>file:hashes.'SHA-256'</code>
<code>ObservableValue</code>	The actual IOC value	<code>198.51.100.42</code> , <code>evil-c2-server.xyz</code>
<code>Pattern</code>	Full STIX detection pattern	<code>[ipv4-addr:value = '198.51.100.42']</code>
<code>Confidence</code>	Indicator confidence (0-100)	<code>85</code>
<code>Data</code>	Full STIX object as dynamic JSON (contains additional context like threat type, description, labels)	<code>{"type": "indicator", ...}</code>
<code>ValidFrom</code> / <code>ValidUntil</code>	Indicator validity time window	Datetime values
<code>IsActive</code>	Whether the indicator is currently active	<code>true</code>
<code>Created</code> / <code>Modified</code>	When the indicator was created/updated	Datetime values
<code>Tags</code>	Sentinel-defined tags	Labels and categories

Tip: Use the `Data` column to extract additional context like threat type and description: `| extend ThreatType = tostring(Data.indicator_types[0])`

Match Indicators Against Your Telemetry

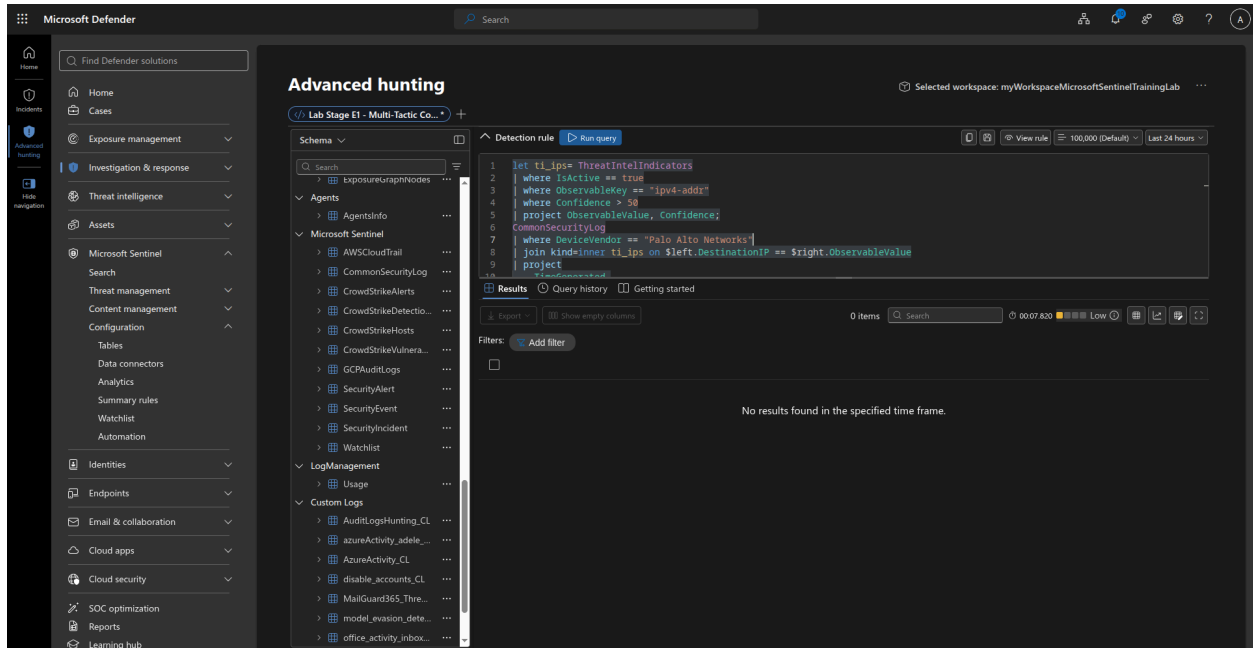
The real power of TI is correlating indicators against your environment's data. Use `join` to find matches between MDTI indicators and your firewall logs:

```
let ti_ips= ThreatIntelIndicators
| where IsActive == true
| where ObservableKey == "ipv4-addr"
| where Confidence > 50
| project ObservableValue, Confidence;
CommonSecurityLog
| where DeviceVendor == "Palo Alto Networks"
```

```

| join kind=inner ti_ips on $left.DestinationIP == $right.ObservableValue
| project
    TimeGenerated,
    SourceIP,
    DestinationIP,
    DeviceAction,
    TI_Confidence = Confidence
| sort by TimeGenerated desc

```

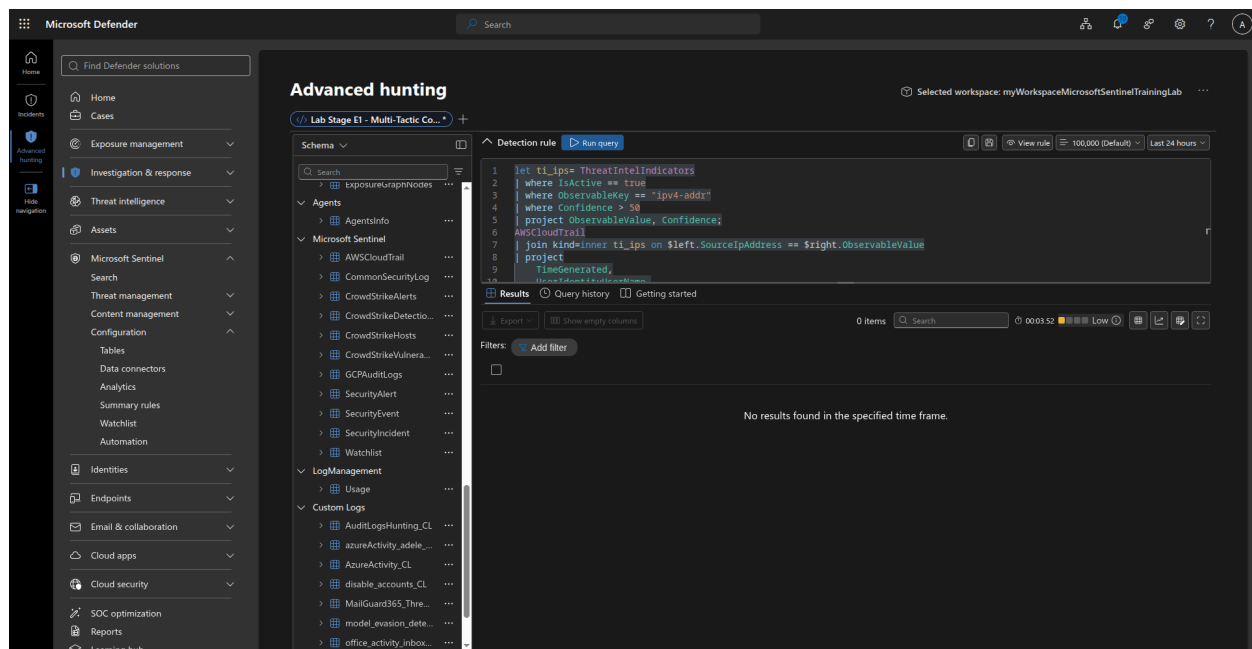


Try the same for AWS CloudTrail:

```

let ti_ips= ThreatIntelIndicators
| where IsActive == true
| where ObservableKey == "ipv4-addr"
| where Confidence > 50
| project ObservableValue, Confidence;
AWSCloudTrail
| join kind=inner ti_ips on $left.SourceIpAddress == $right.ObservableValue
| project
    TimeGenerated,
    UserIdentityUserName,
    EventName,
    SourceIpAddress,
    TI_Confidence = Confidence
| sort by TimeGenerated desc

```



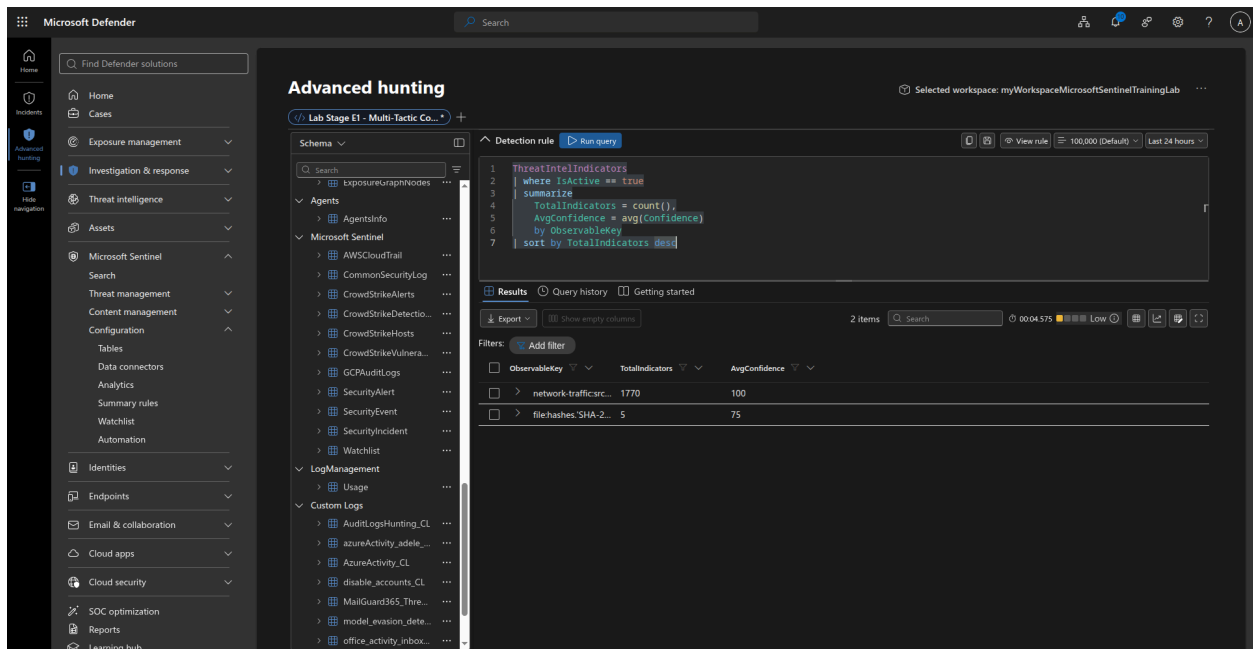
Note: These queries may return **empty results** in the lab environment -- this is expected. The lab's pre-ingested telemetry uses synthetic IP addresses that are unlikely to appear in Microsoft's real-world threat intelligence feed. In a production environment with live traffic, these joins are how you detect connections to known-malicious infrastructure.

What to take away: The query *pattern* is what matters -- `join kind=inner` between a TI indicator table and your telemetry. You can apply this same pattern to any data source (firewall logs, cloud trails, identity events) in your production workspace where real traffic will match real indicators.

Explore Indicator Statistics

Get an overview of your threat intelligence coverage:

```
ThreatIntelIndicators
| where IsActive == true
| summarize
    TotalIndicators = count(),
    AvgConfidence = avg(Confidence)
    by ObservableKey
| sort by TotalIndicators desc
```

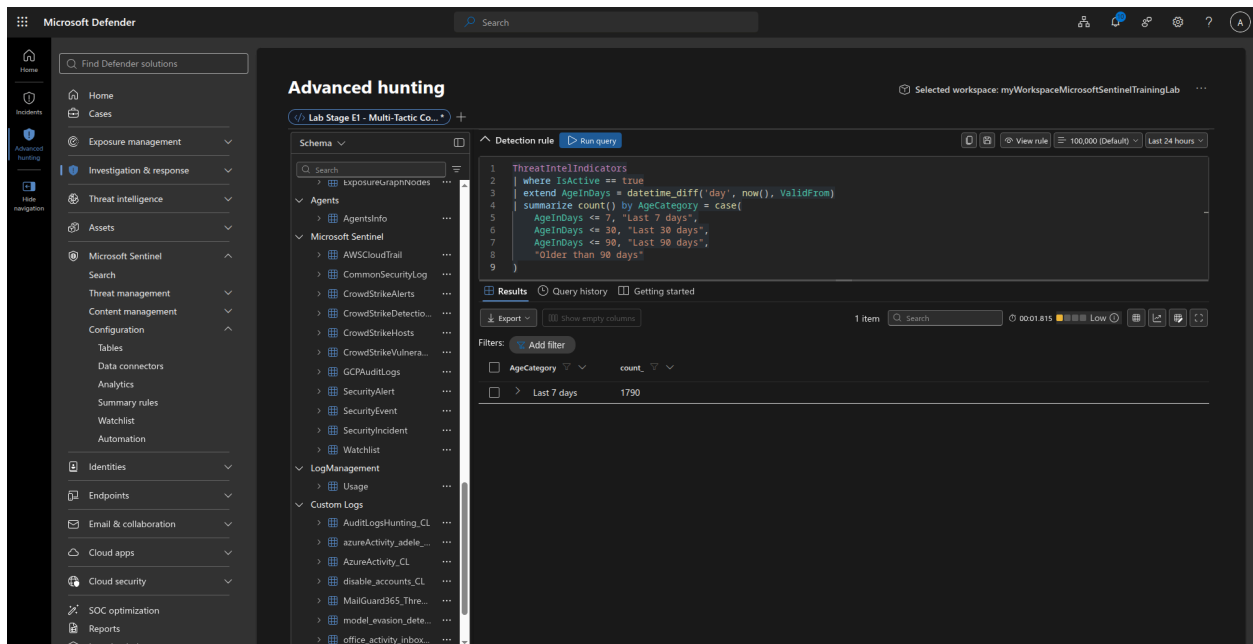


Check the age distribution of your indicators — stale indicators should be reviewed:

```

ThreatIntelIndicators
| where IsActive == true
| extend AgeInDays = datetime_diff('day', now(), ValidFrom)
| summarize count() by AgeCategory = case(
  AgeInDays <= 7, "Last 7 days",
  AgeInDays <= 30, "Last 30 days",
  AgeInDays <= 90, "Last 90 days",
  "Older than 90 days"
)

```



Review the Threat Intelligence Blade

Microsoft Sentinel provides a dedicated UI for managing threat intelligence:

1. Navigate to **Microsoft Sentinel** → **Threat management** → **Threat intelligence**
2. Review the indicators that have been ingested
3. Use the filters to search by indicator type, source, or confidence level
4. Select an indicator to view its full details, including associated threat actors and campaigns

The screenshot shows the Microsoft Defender Threat Intelligence interface. On the left is a navigation pane with options like Home, Incidents, Advanced hunting, and Threat intelligence. The main area displays a table of indicators under the 'Indicators (1,829)' tab. The table has columns for Values, Name, Types, Source, Confidence, and Alerts. The first indicator, 23.144.160.253, is selected. A detailed view of this indicator is shown on the right, including its STIX ID, Name, Types, Pattern, Alerts, Linked cases, Tags, Description, Valid from, Valid until, Kill chains, Indicator types, and Created/Modified dates.

Values	Name	Types	Source	Confidence	Alerts
23.144.160.253	Microsoft Identified IOC	Network traffic	Microsoft Defender Threat...	100	0
213.230.86.245	Microsoft Identified IOC	Network traffic	Microsoft Defender Threat...	100	0
23.234.101.196	Microsoft Identified IOC	Network traffic	Microsoft Defender Threat...	100	0
194.187.176.27	Microsoft Identified IOC	Network traffic	Microsoft Defender Threat...	100	0
199.45.155.66	Microsoft Identified IOC	Network traffic	Microsoft Defender Threat...	100	0
64.62.156.152	Microsoft Identified IOC	Network traffic	Microsoft Defender Threat...	100	0
37.19.221.139	Microsoft Identified IOC	Network traffic	Microsoft Defender Threat...	100	0
45.134.142.211	Microsoft Identified IOC	Network traffic	Microsoft Defender Threat...	100	0
45.168.204.197	Microsoft Identified IOC	Network traffic	Microsoft Defender Threat...	100	0
64.62.156.10	Microsoft Identified IOC	Network traffic	Microsoft Defender Threat...	100	0
155.0.34.22	Microsoft Identified IOC	Network traffic	Microsoft Defender Threat...	100	0
82.226.39.101	Microsoft Identified IOC	Network traffic	Microsoft Defender Threat...	100	0
182.190.221.144	Microsoft Identified IOC	Network traffic	Microsoft Defender Threat...	100	0

Indicator Details:

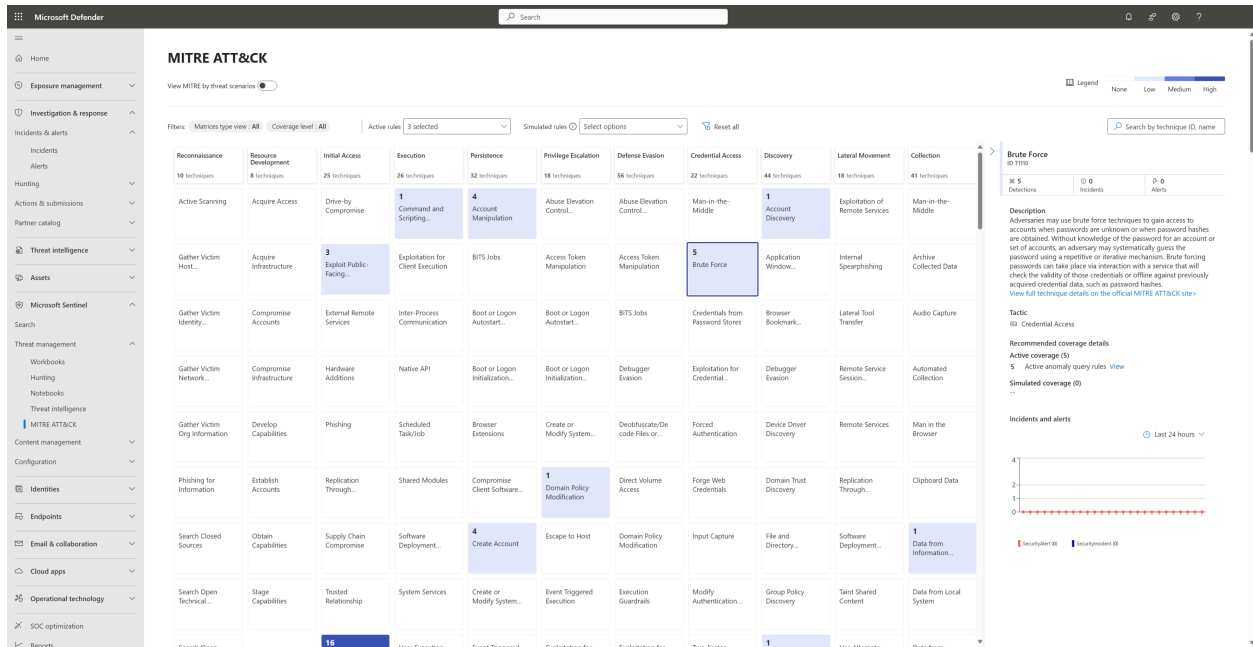
- STIX ID: Indicator--c7ae9be0-e66c-e9ea-1695-8f01699380
- Name: Microsoft Identified IOC
- Types: Network traffic
- Pattern: [network-traffic.src_ref.value = '23.144.160.253']
- Alerts: 0
- Linked cases: Link cases
- Tags: honeypot, pdefault, ic:100, vic:100, +3
- Description: MSTIC Honeypot: An attacker used a brute force attack to gain access to a service or device
- Valid from: 8/17/2026, 3:06:45 PM
- Valid until: 8/17/2026, 8:05:52 PM
- Kill chains: ...
- Indicator types: Botnet
- Created: 8/17/2026, 3:06:45 PM
- Modified: 8/17/2026, 3:06:45 PM
- Created by reference: Identity--c7adab4-c743-4ac3-ac90-798880696a84

Tip: You can also manually add indicators from this blade, upload STIX files, or connect additional TI feeds (e.g., TAXII servers, open-source feeds).

MITRE ATT&CK Coverage

Open the MITRE ATT&CK Page

1. Open the **Microsoft Defender portal**
2. Navigate to **Microsoft Sentinel** → **Threat management** → **MITRE ATT&CK**



You will see the ATT&CK matrix with **colour-coded cells**:

Colour	Meaning
Blue (shaded)	At least one active analytics rule covers this technique
Grey / unshaded	No rule currently covers this technique

The number inside each cell indicates how many rules map to that technique.

Filter by Active Rules

By default, the matrix shows all coverage sources (analytics rules, hunting queries, etc.). To focus on **deployed detection rules only**:

1. Click the **Filters** button at the top
2. Under **Simulated**, deselect all options
3. Under **Active**, ensure **Analytics rules** is selected

MITRE ATT&CK

Workspace: Woodgrove-LogAnalyticsWorkspace

View MITRE by threat scenarios ☐

Legend: None Low Medium High

Filters: Matrices type view : 13 selected Coverage level : All Active rules Active scheduled query rules Simulated rules Select options Reset all Search by technique ID, name

Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access
11 techniques	8 techniques	11 techniques	17 techniques	23 techniques	14 techniques	47 techniques	17 techniques
Active Scanning	Acquire Access	2 Drive-by Compromise	7 Command and Scripting...	29 Account Manipulation	1 Abuse Elevation Control...	Abuse Elevation Control...	Adversary in the Middle
Gather Victim Host...	Acquire Infrastructure	17 Exploit Public-Facing...	5 Exploitation for Client Execution	BITS Jobs	Access Token Manipulation	Access Token Manipulation	24 Brute Force
Gather Victim Identity...	Compromise Accounts	3 External Remote Services	Inter-Process Communication	1 Boot or Logon Autostart...	Boot or Logon Autostart...	BITS Jobs	Credentials for Password Storage
Gather Victim Network...	Compromise Infrastructure	Hardware Additions	Native API	Boot or Logon Initialization...	Boot or Logon Initialization...	Debugger Evasion	2 Exploitation of Credential...
Gather Victim Org Information	Develop Capabilities	3 Phishing	1 Scheduled Task/Job	Software Extensions	Create or Modify System...	2 Deobfuscate/De-code Files or...	Forced Authentication
Phishing for Information	Establish Accounts	Replication Through...	Shared Modules	4 Compromise Host Software...	Domain or Tenant Policy...	Direct Volume Access	Forge Web Credentials
Search Closed Sources	Obtain Capabilities	2 Supply Chain Compromise	2 Software Deployment...	8 Create Account	Escape to Host	Domain or Tenant Policy...	Input Capture
Search Open	Stage	2	2	1	Event Triggered	Execution	1

Command and Scripting Interpreter

ID T1059

7 Detections 0 Incidents 0 Alerts

Description
Adversaries may abuse command and script interpreters to execute commands, scripts, or binaries. These interfaces and languages provide ways of interacting with computer systems and are a common feature across many different platforms. Most systems come with some built-in command-line interface and scripting capabilities, for example, macOS and Linux distributions include some flavor of Unix Shell while Windows installations include the Windows Command Shell and PowerShell.
[View full technique details on the official MITRE ATT&CK site](#)

Tactic
Execution

Recommended coverage details
Active coverage (7)
7 Active scheduled query rules [View](#)
Simulated coverage (0)

Incidents and alerts
Last 24 hours

4
3
2
1
0

SecurityAlert (0) SecurityIncident (0)

The matrix now reflects only your scheduled detection rules. You should see coverage across multiple tactics from the lab's deployed rules.

Explore a Covered Technique

Click on a **shaded cell** — for example, **T1046 (Network Service Discovery)** under the **Discovery** tactic.

A side panel opens showing:

- **Description** of the technique
- List of **analytics rules** mapped to it — you should see rules like **Lab Stage 3.5 - Internal Port Scan Detected (Palo Alto)** and **Lab Stage E2 - Port Scan Detection (Palo Alto)**
- Coverage from **hunting queries** and **threat intelligence** (if applicable)

MITRE ATT&CK > Analytics Refresh

Manage all your rules in one place

Now you can manage all your rules on one page, providing a centralized and streamlined approach to rule management. This enhancement not only simplifies your workflow but also ensures rule location.

353 Active rules

More content at Content hub

Rules by severity

High (120) Medium (257) Low (54) Informational (6)

Active rules Rule templates Anomalies

+ Create Analytics workbooks Rule runs (Preview) Enable Disable Delete Import Export Columns

Search by ID, name, tactic or technique

Techniques : T1059 - Command and Scripting Interpreter Tactics : Execution Rule type : Scheduled Add filter

Severity	Name	Rule type	Status	Tactics	Techniques	Sub techniques	Source name	Last modified
Medium	Office Apps Laun...	S...	Enabled	Executi +2	T1059 +2		Microsoft Defen...	08/04/2024, 21:1...
Medium	Windows Binarie...	S...	Enabled	Execution	T1059		Gallery Content	10/10/2022, 17:2...
Medium	NOBELIUM - Scri...	S...	Enabled	Execution	T1059		Gallery Content	22/03/2021, 02:2...
Low	New CloudShell ...	S...	Enabled	Execution	T1059		Gallery Content	04/03/2021, 01:4...
Medium	Process executed...	S...	Enabled	Executi +1	T1059 +2		Gallery Content	01/05/2020, 19:0...
Medium	Base64 encoded ...	S...	Enabled	Executi +1	T1059 +2		Gallery Content	01/05/2020, 14:2...
Medium	Process executio...	S...	Enabled	Execution	T1059		Gallery Content	01/05/2020, 14:2...

Observation: Notice that some techniques — like **T1046** — have *both* a stage rule (always enabled) and an exercise rule (disabled by default). The stage rules represent the SOC baseline; the exercise rules are for students to practise.

Identify Coverage Gaps

Scroll through the matrix and look for **unshaded cells** in critical tactics. Common gaps you may notice in the lab:

Tactic	Missing techniques	Why
Lateral Movement	T1021 (Remote Services)	No east-west movement data in the current telemetry
Collection	T1560 (Archive Collected Data)	Exfiltration is detected at the network level, not at the archiving stage
Resource Development	T1583 (Acquire Infrastructure)	Pre-attack activity — out of scope for SOC telemetry

Key insight: Coverage gaps do not always mean risk. Some techniques require specific data sources that may not be available (e.g., host-level telemetry for lateral movement). The MITRE page helps prioritise where to invest detection engineering effort.

Map the Lab Attack Chain

Use the MITRE matrix to trace the lab's 10-stage attack chain visually:

- Stage 1 (Initial Access)** — Phishing email delivered via MailGuard → T1566.001, T1566.002
- Stage 2 (Execution)** — Malicious payload executed on endpoint → T1204.002
- Stage 3 (Credential Access)** — Credential dumping via LSASS → T1003.001
- Stage 3.5 (Discovery)** — Internal port scan detected by Palo Alto → T1046
- Stage 4 (Privilege Escalation)** — Okta account takeover with foreign login → T1078.004, T1098

6. **Stage 6 (Exfiltration)** — Large data transfer to external IP → T1041
7. **Stage 7 (Impact)** — AWS IAM escalation and resource abuse → T1078, T1098, T1496, T1562.008
8. **Stage 8 (Persistence)** — AWS backdoor account login → T1136.003, T1078.004
9. **Stage 9 (Privilege Escalation)** — GCP IAM escalation → T1136.003, T1098, T1496, T1562.008
10. **Stage 10 (Initial Access)** — Phishing campaign bypassing email filtering → T1566.001, T1566.002

Each stage corresponds to one or more shaded cells on the matrix. Clicking through them shows the detection rule responsible.

Evaluate a New Detection Idea

Consider a gap you identified in Step 4 and think about how you would close it:

- **What data source do I need?**
- **What KQL query would detect this technique?**
- **What MITRE technique ID should I assign?**

For example, if you wanted to detect **T1110.003 (Password Spraying)** using Okta data:

```
OktaV2_CL
| where TimeGenerated > ago(4h)
| where EventOriginalType == "user.session.start"
| where OriginalOutcomeResult == "FAILURE"
| summarize
    FailedAttempts = count(),
    DistinctUsers = dcount(ActorUsername),
    TargetUsers = make_set(ActorUsername, 25)
    by SrcIpAddr
| where DistinctUsers >= 5
```

This finds a single IP failing authentication against 5 or more distinct users — a classic password spray pattern. If you deployed this as a detection rule and assigned it **T1110.003**, it would appear on the MITRE matrix under **Credential Access**.

Tip: You do not need to deploy this rule. The point is to understand how the MITRE coverage page reflects your detection engineering decisions.

Remaining exercises after this one:

- **Exercise 5** — Cross-Platform Response Actions: Device Isolation (*optional, requires MDE*)
- **Exercise 6** — Port Scan Detection & Threshold Tuning
- **Exercise 7** — Okta MFA Factor Manipulation
- **Exercise 8** — Watchlist Integration
- **Exercise 9** — Cost Management & Ingestion Analysis
- **Exercise 10** — Table Management: Tiers & Retention
- **Exercise 11** — Data Lake KQL Jobs
- **Exercise 12** — Data Lake vs Real-Time Detection
- **Exercise 13** — Data Lake Notebooks
- **Exercise 14** — Sentinel MCP Server Demo Prompts
- **Exercise 15** — Data Federation with ADLS Gen2

- **Exercise 16** — Data Transformation: Split Ingestion by Tier
- **Exercise 17** — Custom Graph: Cross-Source Attack Chain